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**Ethical Foresight: Confronting Misinformation, Representation and Toxicity in
Generative AI**

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ABSTRACT

This thesis investigates the socio-technical harms posed by generative AI, specifically focusing on misinformation, representational errors, and toxicity in LLMs. By critically examining existing safety evaluations within these domains, the research aims to develop a comprehensive framework to address ethical risks in AI.

The study systematically reviews the current landscape of socio-technical safety evaluations to construct a taxonomy that can be applied during the development of LLMs. It identifies how these harms manifest in LLMs, the mechanisms through which AI systems propagate misinformation, errors, and toxicity, and how these factors contribute to social inequities.

The findings emphasise the critical need for interdisciplinary approaches to reform generative AI development practices, ensuring these technologies are socially responsible and beneficial to all. This research advances the discourse on AI ethics and safety evaluations, providing actionable insights for developers, policymakers, and stakeholders to promote more equitable AI systems. The study's insights can guide the creation of robust ethical compliance frameworks, ensuring AI deployment aligns with societal values and legal standards.

Key Words: AI ethics, Sociotechnical harms, Misinformation, Representation, Toxicity, Safety evaluations

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INTRODUCTION

Artificial Intelligence (AI) is transforming our global landscape, and comprehending its ethical implications extends beyond the responsibility of developers to become a universal mandate. It is increasingly being recognized that the ethical dilemmas and challenges associated with AI are among the most critical issues confronting our era.

Generative AI, a subset of AI, has revolutionised content creation across various domains including art, music, literature, and design. It encompasses algorithms and models capable of generating content that mimics human creativity, often indistinguishable from human-generated content. This ability has opened up new avenues for innovation and creativity, but it also raises profound ethical questions regarding its societal impact. It extracts insights from large datasets, identifies patterns, and makes predictions or judgments, and industries have become reliant on them. It has led to an increase in productivity by automating operations, optimising workflows, and lowering manual labour (ibid). This dependence on AI extends to fields including marketing, banking, cybersecurity, healthcare, and transportation, and those that don't adapt will quickly be left behind (Aldoseri, 2023).

Language models can be thought of as linguistic manipulators, as they are optimised to mirror natural language patterns, detecting statistical trends present in their training data (Weidinger et al, 2021). Although this is not inherently negative it becomes problematic when the training data is biased, discriminatory, or toxic (Shah et al., 2020). This is because LLMs are a reflection of human values, thoughts and they mirror reality. People can decontextualize certain aspects of reality and recontextualize them in different ways using different languages but since no language can ever completely separate itself from the social reality in which it is rooted, it is a process rife with bias (Hackett, 1984). Not only is language a biased spectator, but it is limited by its own structure to a point where it may exclude opposing viewpoints and interpretations in the interest of upholding its own values and beliefs (Paul and Elder, 2020). Understanding bias requires an appreciation of the intrinsic and inescapable impartiality of language (Burkholder et al, 2022).

Based on historical data, algorithms also predict patterns, behaviours, or consequences (ibid). Sales forecasting, weather forecasting, recommendation engines in streaming services, e-commerce platforms, and social media all make use of this capacity, which offers tailored

content or product recommendations based on user preferences (Aldoseri, 2023). These historical biases in training data, gathered from contexts of systemic injustice, lead to underrepresentation of certain communities in the data and models that fail to accurately represent marginalised groups, which is known as the "undersampled majority" (Raji, 2020). This in turn leads to "bias amplification" as LLMs designed to accurately reflect language, they overrepresent even little biases in the data (Wang and Russakovsky, 2021; Zhao et al., 2017). AI generated misinformation has been weaponized as a tool used for propaganda, control and manipulation. The business of misinformation has exploded with the advent of generative AI, as it has never been this easy to manipulate the truth. As misinformation rises rapidly, and tech progresses on the side, there is a dire need for AI governance to ensure that there are safety mechanisms in place. Research has shown that AI systems can spread false information, undermine labor and employment, exhibit algorithmic bias related to age, gender, ethnicity, and disability, and sustain structural injustices (Ho et al, 2021). It is difficult to comprehend the risks that generative AI could pose, especially considering how rapidly multi-modal generations can be made, how similar they are to human made outputs, and how adept they are at adapting. It is important to curb risks to ensure that the implications it has on society is mitigated. Especially since these issues are perpetuated at a much faster rate due to the large dissemination powers of AI. Secondly, mitigating ethical risks allow for advancement of scientific growth and technological advancement since better regulated LLMs will help the world grow without it posing a threat to humanity.

There is great effort being put into the field of computer science to develop technical bias definitions and fairness metrics as well as bias detection and mitigation strategies, but it is difficult to implement these by design (Deck et al, 2024). These questions are hard to reach a consensus on; How should we ensure AI treats everyone equally? Who is responsible if the AI makes a mistake? How can we make AI decisions understandable to everyone? How do we protect people's personal information? This makes the normative challenge a central issue in the development and deployment of AI systems (Barocas & Narayanan, 2019). Furthermore, the discussion on algorithmic fairness is often limited to academic circles and relies on self-regulation. This can lead to problems in ensuring that fairness principles are actually applied and enforced. In the words of Dr. Joy Buolamwini, "The rising frontier for civil rights will require algorithmic justice. AI should be for the people and by the people, not just the privileged few. Data is a function of our history, so the past dwells within our algorithms. It is showing us the inequalities that have always been there" (Hamilton, 2020).

Relevance to Security Studies

The relevance of this research to the field of Security Studies lies in its focus on the intersection of technical and social dimensions of AI safety. Security Studies traditionally encompasses the protection of national security, human rights, and societal well-being. With the increasing integration of AI into critical aspects of these pillars of society, understanding the sociotechnical harms is imperative for ensuring comprehensive security.

The research underscores a fundamental struggle in AI evaluation: balancing the scope of what to evaluate with the need for comprehensive coverage. Current evaluation methods often face the dilemma of being either too broad to capture nuanced harms or too narrow to be broadly applicable. This dichotomy is particularly challenging in the context of LLMs, where the vastness of possible outputs and interactions makes it difficult to ensure comprehensive ethical safeguards.

The analysis will demonstrate that progress in AI safety cannot be achieved through technical means alone. While algorithmic improvements and automated testing are essential, they are insufficient without informed human judgement and oversight. This includes precisely defining expected types of biases and the contexts in which they might emerge. Human oversight ensures that evaluations consider the subtleties and complexities of social impacts, which purely technical approaches might overlook. To bridge this gap, the research proposes a taxonomy to categorise harms and analyse ethical considerations while developing LLMs. Developing taxonomy serves the several critical functions of establishing a common language and criteria for evaluating AI systems, facilitating more consistent and comparable assessments. This helps inform policymakers and ethical boards, aiding in the creation of guidelines that address specific sociotechnical harms associated with AI.

Safety Evaluations Repository- Google DeepMind

The Sociotechnical Safety Evaluation repository, developed by DeepMind, is a critical resource for evaluating the ethical and social risks associated with generative AI systems. This extensive repository was used as the primary source for evaluating these ethical risks, guiding the development of strategies that encompass technical, social, and systemic dimensions. This analysis marries philosophical, sociological, and sociotechnical understanding of AI. It is

necessary to clearly define each category as this is a constantly evolving field that needs to be defined for it to be regulated.

The repository is an expansive collecting archive that is updated periodically by researchers and contributors from various institutions, focusing on the latest developments and findings in AI safety and ethics. The repository is updated to include the most recent research, case studies, and evaluation frameworks. Additionally, the repository is open-access and in the public domain, allowing even small time-researchers or tech enthusiasts to contribute to this community-sourced repository of safety evaluations. This distinction is of paramount importance as it allows for democratically sourced lists where key tenets like transparency and equity are reinforced. The frequency of updates ensures that the repository remains a relevant and comprehensive resource for ongoing research and policy-making. The repository provides a structured, three-layered framework that assesses AI systems' capabilities, human interaction, and systemic impacts. This multi-faceted approach ensures a thorough evaluation of the potential risks and benefits of AI technologies (Weidinger et al., 2023). By mapping out the current state of AI safety evaluations, the repository helps identify significant gaps in existing methodologies. It highlights areas such as contextual risks, specific risk evaluations, and multimodal risks, guiding future research and policy interventions (ibid). The framework emphasises the shared responsibility between AI developers, application developers, public authorities, and the broader public (ibid).

LITERATURE REVIEW

A. Underlying Causes of Harms

There are many definitions of bias that differ based on the domain; law, psychology, data science, legal, healthcare, etc. that depend on the context, culture, and research fields (Yogarajan et al, 2023). In various disciplines, the core principle of addressing bias involves assessing the harm it causes but in LLMs it is difficult to pin-point the origin of the harm or the effect of the harm it causes. Bias in LLMs is not intrinsic or intentional but stems from unequal treatment or outcomes among social groups that could be due to a variety of issues like misrepresentation, stereotyping, uneven system performance, and both direct and indirect discrimination.

In society it is easy to find the root of these targeted beliefs, it is directed at a set of attributes like gender, age, race, and other identity markers. This cannot be specified in LLMs so the task of detecting, quantifying, and evaluating bias in these models is complex. It is critical to define it to be able to regulate and mitigate bias and ensure fairness and equity in decision-making processes and outcomes. To estimate and anticipate their widespread use and potential impact on diverse populations it would help to narrow it based on: Data Bias, Algorithmic Bias, Interaction Bias and Systemic Bias.

Data Bias:

1. Training Data Imbalance:

Underrepresentation: If certain demographics are underrepresented in the training data, the model may perform poorly for these groups. **Overrepresentation:** Conversely, if the training data disproportionately represents certain groups, the model might overfit to these characteristics, leading to biased outputs favouring those groups.

For example, facial recognition systems often have higher error rates for people with darker skin tones due to the lack of diverse training data. Studies have shown that without a dataset that has labels for diverse skin types, including amount of hair, thickness, colour, the accuracy of automated skin cancer detection systems lessens (Esteva et al., 2017). This has implications in the field of healthcare, as well as in law enforcement, as it was revealed that African Americans are more likely to be subjected to frequent targeting and face recognition searches (Garvie et al, 2016).

2. Historical Bias:

Pre-existing Societal Bias: Training data that reflects societal biases (e.g., gender or racial biases in hiring practices) will carry these biases into the model, perpetuating discriminatory patterns. **Sampling Bias:** The way data is collected can introduce bias if certain populations are systematically excluded or included more frequently than others.

Word embeddings, a foundational technique in NLP captures semantic relationships between words by placing similar words close together in a high-dimensional vector space (Bolukasi et al, 2016). The vector representations of words exist in a continuous vector space where words that have similar meanings have similar representations (ibid). Word2Vec, GloVe and LSTM

models such as context2vec and ELMo, are some of the popular methods for creating these embeddings, using neural network models to learn word embeddings from a large corpus of text (Bender et al, 2021). As a result, bias, toxicity and misinformation in the training data, such as hate speech, profanities, or offensive language, is learned by the model and represented in the word embeddings (Bolukasi et al, 2016). These embeddings then contribute to the generation of toxic content. Word2Vec completed the missing word of the analogy “man is to computer programmer as woman is to ‘X’ with the word “homemaker” (ibid). Word embeddings capture word meanings based on their co-occurrence patterns but often lack deeper contextual understanding (Buolamwini et al, 2018). The phrase "You throw like a girl" might be interpreted as merely a reference to throwing, missing the sexist and derogatory connotation (ibid). The extent of bias is such that prediction models have shown that the activity “cooking” is over 33% more likely to involve females than males in a training set, and this bias is amplified by trained models to 68% at test time (Zhao et al, 2017).

Algorithmic Bias:

1. **Model Architecture and Design:** The design of neural networks might emphasise features that are more prominent or easier to learn, leading to biased representations. For instance, certain architectures might overemphasise text patterns from specific dialects or sociolects when processing natural language, which can result in biased language models that underperform on text from less represented dialects (Mehrabi et al, 2021). The choice of activation functions and how they process input data can also contribute to bias if they amplify certain patterns over others. It is used in a variety of applications; childhood welfare systems, autonomous vehicles, employment matching, automated legal aid for immigration algorithms, flight routing, search and advertising placement algorithms (ibid). This signifies the importance of protecting downstream applications from harmful effects when modelling an algorithm. In autonomous vehicles, for instance, a model might be trained predominantly on data from urban environments, causing it to perform poorly in rural settings (Yusuf et al, 2024). This can lead to safety issues as the vehicle may not correctly interpret or respond to rural road conditions, which are less represented in its training data (ibid).
2. **Feature Extraction Bias:** Features extracted during training might carry inherent biases if they disproportionately represent specific patterns. Deep learning models that automatically learn features from data can inadvertently learn biased representations if the training data is biased. In healthcare, a machine learning model for diagnosing

diseases might learn to associate certain medical conditions with high-risk labels based on biased training data (Nazer et al., 2023). If the training data disproportionately represents older patients, the model might inaccurately predict higher risks for older demographics while underestimating risks for younger patients with similar conditions (ibid). This type of bias can significantly affect diagnostic accuracy and treatment recommendations (ibid). Furthermore, high-dimensional data analysis raises serious concerns about feature selection bias, especially in biological applications (Krawczuk & Lukaszuk, 2016). It can result in overfitting and worse generalisation performance when the same dataset is used for both feature selection and model learning (ibid). Researchers have suggested a number of methods to deal with this problem, such as employing different datasets for selection and learning (Krawczuk & Lukaszuk, 2016) and theoretical frameworks for estimating true selection error (Raudys et al., 2005). In high-impact applications, constructing dependable and trustworthy machine learning models requires both understanding and mitigating feature selection bias.

Interaction Bias:

1. **Reinforcement Learning from Human Feedback (RLHF):** RLHF is a technique used to train AI models, especially those involved in complex decision-making tasks, by incorporating human feedback into the learning process. This method combines elements of reinforcement learning (RL) with human-in-the-loop training to guide the model's behaviour based on human preferences and judgments (Christiano et al., 2017). Although human oversight is an integral part of training, models can become biased if the feedback itself is biased. If users consistently provide positive feedback for biased outputs, the model will learn to favour those outputs (ibid). AI systems that adapt based on user interactions can develop biases if they encounter biased user behaviour, such as repeated exposure to discriminatory language or patterns.
2. **Adversarial Manipulation:** Adversaries can exploit known biases in models to manipulate outcomes. For instance, Microsoft's chatbot Tay had to be taken down when hostile users provoked it to post offensive, sexist and racist tweets to more than 50,000 followers (Lee, 2016). The AI was designed to mimic the behaviour of humans, but failed dramatically in recognizing potential toxic behaviours (Biggio et al, 2018). ML systems are designed to deal with "known unknowns"; situations where the possible outcomes are understood, but the exact occurrences are uncertain (ibid). However, adversarial machine learning operates in a more challenging environment where it must contend

with "unknown unknowns" (ibid). This unpredictability leads ML systems to misclassify these new inputs with high confidence, which pose serious security threats (ibid).

Systemic Bias:

1. **Deployment Context:** The deployment context of AI is crucial in determining its effectiveness, fairness, and overall impact. Context encompasses the environment, user base, operational parameters, and societal implications in which the AI system functions. A prime example is when Porcha Woodruff, an eight months pregnant black woman was wrongfully arrested for carjacking due to faulty AI identification in Detroit (Bhuiyan, 2023). AI, specifically facial recognition software like Clearview AI and Amazon's Rekognition, has led to harmful consequences for underrepresented and marginalised communities as biased algorithms disproportionately impact people of colour, exacerbating systemic discrimination, and causing significant personal and social harm (ibid). To mitigate these issues, the deployment context must be carefully considered, shifting the focus from merely predicting crime to understanding the underlying factors contributing to criminal behaviour (Barabas et al, 2018). These factors can then be analysed in a causal model to understand the social, structural, and psychological reasons behind criminal behaviour to support risk mitigation, without relying on flawed risk predictions (ibid). Bias can also be amplified through feedback loops where the AI system's biased outputs influence future data, further entrenching the bias (ibid).
2. **Decision-Making Framework:** Decisions about thresholds and cutoffs for classification tasks can introduce bias if they disproportionately affect certain groups. Adjusting these thresholds based on biased training data can lead to unfair outcomes. Lack of transparency; interpretability and explainability in how decisions are made can obscure underlying biases, making it difficult to identify and mitigate them. Taking the example of credit scoring, if the threshold for approving a loan is set based on biased training data, it may unfairly disadvantage minority groups, leading to lower approval rates for these populations. Thresholds often involve a trade-off between sensitivity (true positive rate) and specificity (true negative rate), and so a threshold optimised for overall accuracy may not be fair to all groups (Barocas & Narayanan, 2019). In a medical diagnosis model, optimising the threshold for overall accuracy might result in lower sensitivity for a minority group, meaning they are less likely to receive correct diagnoses (ibid). This lack of clarity can result in persistent unfair outcomes, as biases remain hidden within the decision-making algorithms (ibid).

B. Manifestation of Harms

As Andrej Karpathy puts it “Hallucination is all LLMs do, they are dream machines. We direct their dreams with prompts. The prompts start the dream, and based on the LLMs hazy recollection of its training documents, most of the time the result goes someplace useful. It's only when the dreams go into deemed factually incorrect territory that we label it a ‘hallucination’. It looks like a bug, but it's just the LLM doing what it always does.” LLMs can generate outputs that are plausible-sounding but factually incorrect or nonsensical. This occurs because LLMs are designed to predict the next word in a sequence based on patterns in the training data, without truly understanding the content. Bias in data is a major reason why LLMs generate biased, toxic, and misleading content. Training data often reflects the societal and historical biases present at the time of data collection. This includes biases related to gender, race, ethnicity, nationality, sexuality, and disability. The methods and sources used for data collection can also introduce bias. Oftentimes this manifests in the form of implicit bias, or the unconscious associations and attitudes that shape our perceptions and actions without our explicit awareness. Implicit bias can manifest through the data used to train models, the features selected for analysis, and the classification rules established for decision-making processes (Wachter, Mittelstadt, & Russell, 2020). Unlike explicit biases, which are conscious and deliberate, implicit biases are subtle and pervasive, often operating below the level of conscious thought. This is why LLMs are often referred to as “stochastic parrots” because their “knowledge” is purely syntactic, and they do not possess consciousness or intentionality (Bender et al, 2021).

AI systems' ability to process vast amounts of data and detect patterns that may not be immediately apparent to human analysts can lead it to uncover relationships and connections that might seem counterintuitive. This same capability can also lead to the emergence of implicit biases within AI systems. They operate subtly, shaping our thoughts and actions in ways that perpetuate systemic discrimination even in the absence of overt prejudice (Galvan & Payne, 2024). Coded language and caustic remarks can propagate harmful stereotypes and prejudices without being overtly recognized as hate speech, thus continuing to inflict damage under the radar (Yoder et al, 2022). Identifying and mitigating implicit bias requires an understanding of the nuanced ways in which it manifests, including subtleties, innuendos, and cultural references that purely technical models might overlook (ibid).

Furthermore, data scraped from the internet often includes unmoderated content, which may be biased or toxic. This is compounded by the fact that some communities are underrepresented online, leading to skewed representations in the data (Zewe, 2022). If a dataset contains predominantly western-centric content, the model will likely favour western perspectives and norms (Hartmann et al., 2023). During the training process, models may overfit to biased patterns, reinforcing and even exacerbating these biases in their outputs (Fang et al, 2023). It is crucial to understand how these biases manifest as different types of ethical harms. However, it is important to note that these harms are not only because of flawed data, but can also be manifested as a result of technical issues in the LLM:

1. **Contra-factual Bias:** This occurs when LLMs assume the truth of a false premise presented in a query. This issue is exacerbated by the models' tendency to prioritise generating responses over validating the content. For example, if asked a question based on an incorrect fact, the model may generate a detailed but inaccurate response without verifying the premise (Dahl et al, 2024). This problem is exacerbated by the instruction-following nature of many LLMs, which prioritise responding to the user's query over validating the content.

2. **Error Propagation:** LLMs often struggle with self-correction and error propagation. When an error is made, it can compound in subsequent responses. While some methods like self-rethinking frameworks aim to mitigate this by allowing models to re-evaluate their answers, these solutions are still in developmental stages. If an LLM incorrectly states a historical fact, such as misidentifying the year of a major event, subsequent queries based on this misinformation will propagate the error, leading to increasingly inaccurate outputs (Jiang et al., 2023; Huang et al., 2024).

3. **Model Overconfidence:** LLMs can exhibit overconfidence, where the model's certainty in its responses does not correlate with the accuracy of those responses. Larger and more instruction-tuned models tend to show higher levels of overconfidence, leading to an increased likelihood of presenting incorrect information as factually accurate (Duan et al., 2023).

4. **Complexity of Errors:** The errors made by LLMs are often intricate and diverse, spanning across different domains such as arithmetic reasoning, commonsense reasoning, and linguistic context errors. This complexity makes it challenging to generalise fixes or improvements, as the nature of errors can vary significantly with different tasks (Li et al., 2023). This complexity makes it challenging to generalise solutions across different types of errors.

5. Performance Variability: LLMs exhibit performance variability across different types of queries and contexts. They tend to perform better with well-documented information but struggle with less prominent or older data, leading to inconsistent accuracy and reliability. LLMs perform variably across different types of queries and contexts. They tend to perform better with well-documented and frequently cited information, such as Supreme Court cases, but struggle with less prominent or older cases (Dahl et al, 2024). As not all data is available like this, the variability can lead to inconsistent accuracy and reliability, depending on the context of the query (ibid).

These technical manifestations point to the importance of 'Alignment'; the challenge of ensuring that AI systems meet the expectations of individuals and groups, while operating in diverse contexts and having their own inherent uncertainties. Bruno Latour's Actor-Network Theory suggests that both human and non-human entities should be considered equal participants in a network, which challenges traditional hierarchical structures through network analysis (Li & Zhu, 2024). Giving human and non-human actors equal consideration at the outcome assessment would help foresee the implications of AI (ibid). Ensuring fairness, accountability, transparency, and interpretability (FATI) in proper governance frameworks and understanding the interactions between actors in the network are essential to foresee and mitigate the implications of AI (Li & Zhu, 2024). Technology development, marketing, customer acceptance, market penetration, and technological evaluation and iteration are all integral features of generative AI that depend on three main pillars, capital, human resources, and technology (ibid). ANT necessitates understanding the actors, their translations, and the complex connections between them to comprehend the impact and the formation of ethical difficulties (ibid).

These factors underscore the need for continuous refinement in the development and evaluation of LLMs, as well as the importance of integrating robust error-detection and correction mechanisms to improve their reliability and trustworthiness. On the flip side, AI/ML technologies offer substantial promise in fighting misinformation due to automation which reduces the costs by optimising resources and decreasing reliance on human labour (Takyar, 2023). This efficiency also benefits resource-limited organisations by maximising the reach and impact of their messages (ibid). These systems continuously improve through machine learning, adapting to evolving misinformation tactics. Additionally, AI/ML enhances media literacy by categorising information sources and offering insights into content credibility, helping the public become more discerning consumers of information (Jones-Jang & Mortensen, 2019). This

fosters a more informed and critical public, reducing the spread of false information and enhancing the overall quality of the information ecosystem. In the upcoming sections, I will delve more deeply into the individual aspects of Misinformation, Representation and Toxicity.

C. Misinformation

Misinformation is defined as false or partially false information which can be spread both unintentionally and intentionally” (Zhou et al, 2023). For the purpose of this research, the term "misinformation" will be used to describe false or partially false information propagated by LLMs, regardless of the intention behind it. Unlike terms such as "disinformation" or "fake news," which imply a deliberate intent to deceive, "misinformation" does not carry this connotation. The concept of *mens rea*, Latin for "guilty mind," highlights the importance of intent in distinguishing between these terms. However, when it comes to LLMs, attributing intent is problematic, as these models generate and spread information without a conscious motive. Consequently, the term "misinformation" is more appropriate, as it encompasses both intentional and unintentional dissemination of false information.

Misinformation has played a major role in pivotal moments in history, from communication strategies of Roman emperors who utilized coins to influence the masses, to the propaganda machinery of Nazi Germany, which heavily relied on printed media, radio broadcasts, and cinema (Barman et al, 2024). The advent of digital platforms and the internet has exponentially amplified the reach and impact of disinformation campaigns. Following the 2016 US presidential election, the term "fake news" gained considerable traction within both the media and academic domains (ibid). This concept, akin to misinformation, encompasses narratives, theories, or opinions that swiftly proliferate across social media or online channels.

Building on the work of Wardle (2017) misinformation can be thought of as information that is;

- **Misheard or Misquoted:** Genuine news that is distorted through mishearing or misquoting. This often occurs when information is relayed through multiple sources, each potentially altering the original message.
- **Fabricated to make profit:** Made-up content, usually sensational stories intended to attract clicks or generate ad revenue. These stories are designed to be attention-grabbing to maximize their viral potential.

- **Promoting a person, party, or ideology:** Pieces are created with the intent of influencing public opinion or political outcomes. They may include exaggerations, false claims, or selectively presented facts.
- **Jokes taken as fact:** Satirical content intended for humor gets misinterpreted as genuine news. This often happens when the audience is not familiar with the satirical nature of the source.
- **False connection or False content:** Headlines, captions or visuals do not match the content.
- **Manipulated or Imposter content:** Genuine information that is manipulated or true sources that are impersonated.

This misinformation finds its way to be masked as truth through various sophisticated and often insidious mechanisms; through word-of-mouth, social media, and re-reporting by other media outlets, often without verifying the original context (ibid). These mechanisms are multifaceted and pervasive, and are known to exploit human cognitive biases, the dynamics of social networks, and the structure of media systems. They spread rapidly through clickbait headlines, misleading thumbnails, and shared links, exploiting the monetization models of online advertising (ibid). They employ targeted social media campaigns, political ads, and partisan news sites, often using bots and trolls to amplify reach.

Generative AI has been exploited by various actors, including activist or political organizations, bots, criminal and terrorist organisations, and even governments (Barman et al, 2024). These entities leverage AI to create and disseminate false narratives, manipulate public opinion, and advance their agendas. The ease of content creation, coupled with the rapid dissemination facilitated by digital platforms, has heightened the effectiveness of such campaigns. Moreover, the ability to micro-target specific audiences has further intensified their impact (ibid). For instance, activist groups may use generative AI to produce convincing but fabricated evidence to support their claims, such as spreading false information about the safety or efficacy of vaccines (ibid). Bots, operating as part of botnets, also utilise AI to spread misleading content across online platforms, amplifying the reach of false narratives (ibid). Criminal and terrorist organisations employ generative AI to craft persuasive propaganda materials aimed at recruiting individuals or sowing discord within communities (ibid). Governments, both domestic and foreign, harness generative AI to influence public opinion on critical issues, often disseminating false narratives through social media channels (ibid).

From a security studies perspective, the ability of LLMs to generate and disseminate misinformation also poses a significant threat to national security institutions. This capacity to generate misinformation on a large scale, poses a threat to the pillars of democracy: democratic representation, accountability, and societal trust (Kreps & Kriner, 2023). Involving the public in the political process has been the backbone of democratic institutions, but with the advancement of LLMs there's a risk that generative AI could be misused to push specific agenda. This manipulation of information landscapes can impair the capacity of security institutions to effectively address genuine threats, as their efforts may be diverted or undermined by false narratives and artificially generated public sentiments (Helmus & Chandra, 2024). A notable example of astroturfing occurred in 2017 when bots inundated the Federal Communications Commission with over eight million comments advocating for net neutrality repeal (ibid). However, these comments were largely identical, indicating a coordinated effort to deceive rather than genuine grassroots support. With advancements in AI, detecting such manipulation becomes increasingly challenging, making it hard for agencies to discern authentic stakeholder preferences (ibid). Moreover, adversarial actors could leverage AI-generated misinformation to conduct information warfare, disrupt social cohesion, and manipulate political processes, thereby compromising the integrity and stability of democratic societies (ibid). Newsguard, a misinformation analytics company has catalogued 831+ websites disseminating Unreliable AI-Generated News (UAINS) encompassing 16 languages¹ (Brewster & Sadeghi, 2023). These platforms have found to bear generic monikers such as 'iBusiness Day', 'Ireland Top News', and 'Daily Time Update', which primarily publish articles authored by artificial intelligence, covering a variety of topics including politics, technology, entertainment, and travel (ibid). These sites function with minimal human supervision and contain false baseless assertions, including news about political figures, celebrity death hoaxes, invented events, and articles that present past events as recent occurrences (ibid). The nature of the industry ensures that leading brands and outlets support these articles as these websites employ a revenue model based on programmatic advertising, wherein the ad-tech industry indiscriminately delivers advertisements, regardless of the website's nature or quality (ibid). This content can be extremely misleading for consumers who are tricked into believing they are established news outlets.

¹ Arabic, Chinese, Czech, Dutch, English, French, German, Indonesian, Italian, Korean, Portuguese, Russian, Spanish, Tagalog, Thai, and Turkish

An example of a significant misinformation campaign or influence operation was the “Spamouflage” operation, a Chinese state-driven initiative, used AI to create misinformation against China’s adversaries while promoting its global image. Coordinated by the Chinese Communist Party (CCP), Meta recently uncovered and removed numerous social media accounts involved in this campaign (Bergengruen, 2023). This AI-generated content circulates within echo chambers, receiving significant engagement to flood discussions with misinformation (Flaxman & Rao, 2016). Techniques include anti-West narratives, criticism of regional adversaries, and ridicule of CCP critics through spoofed social media and explicit images (ibid). Realistic fake profiles were created using Generative Adversarial Networks (GANs) to deceive users. The operation employed deep fakes such as the fictitious “Wolf News” with deepfake TV anchors spreading misleading content about the US (Graphika, 2023).

It is important to note that human generated misinformation differs from AI- generated misinformation in some explicit patterns. A study showed that there were four specific linguistic differences in AI misinformation, namely their tendency to: improve details, convey doubts and constraints, make deductions, and mimic individual tones (Zhou et al, 2023). Added to this, existing detection models are shown to perform poorly when it comes to AI misinformation. The other dominant solution to misinformation being used in industry is information assessment guidelines which again performs poorly as AI misinformation usually adheres to assessment guidelines of being transparent, comprehensive and credible (ibid).

These problems have manifested in various ways, often resulting in generations that have caused real-life harm. Covid 19 was one of the first instances where AI generated content posed a significant risk. In 2020, when asked questions about possible cures, OpenAI's GPT-3 produced inaccurate results regarding COVID-19 treatments (Spitale et al, 2023). Although the generated comments lacked scientific validity, they were extensively shared on social media platforms, aiding in the dissemination of false information in the pandemic (ibid). This points to the phenomenon that when people are overwhelmed with too much information that's hard to understand or trust, they tend to give up trying to evaluate it critically (ibid). This can make them reliant on simple cues or heuristics which makes it all the more critical to have safeguards to prevent harmful effects of misinformation (ibid).

D. Representation

Representational harms occur when algorithmic systems reinforce and amplify societal inequalities through biased representations. These harms disproportionately affect marginalised communities, perpetuating stereotypes and social hierarchies (Angwin et al., 2016; Chouldechova, 2017).

Representational harms are said to manifest through various means; **Inequitable Distribution of Resources**: Systems disproportionately favour certain groups, leading to unequal access to resources (Dressel & Farid, 2018). **Hierarchical Representations**: Depictions that reinforce social hierarchies, often based on race, gender, or disability (van Es et al., 2020). **Disparate Performance**: Algorithmic systems perform worse for certain identity categories, exacerbating inequalities (Dieterich et al., 2016). **Entrenchment of Inequalities**: Social and economic disparities are deepened by biased algorithmic decisions (Kleinberg et al., 2016).

Bias and representation harms can occur at multiple stages in the LLM lifecycle, as each part affects it differently. These include the datasets used for training, evaluating, auditing, and benchmarking models, which may inherently contain biases (Bergman et al, 2023). Additionally, the stakeholders who evaluate the technology and the annotators who label datasets or probe models can introduce their own biases. Even the characteristics and perspectives of the ML designers themselves can influence the fairness of the model (ibid). These harms occur through the interplay of technical components, such as biased datasets and algorithms, and societal power dynamics that define what is considered "normal" or "standard" (Angwin et al., 2016). AI communication that positions one group as superior or reinforces negative stereotypes about another group can subtly disadvantage marginalised populations, especially in educational settings (Dressel & Farid, 2018).

These harms emerge from the interplay of technical and social factors as algorithms often adopt societal norms and power structures, embedding systemic biases (Shelby et al , 2023).

Under-diagnosed patient populations were consistently and selectively underdiagnosed by classifiers created using cutting-edge computer vision techniques, and the underdiagnosis rate was higher for intersectional under-served subpopulations, such as Hispanic female patients (ibid). The use of AI systems in medical imaging for disease diagnosis that have such biases raises ethical questions about the usage of these models in clinical settings since they run the

danger of aggravating pre-existing care biases and possibly creating uneven access to medical care (Seyyed- Kalantari, 2021).

Predictive policing is an example that shows us how unfair algorithms can have real life implications. It has shifted the focus from past pattern analysis to forecasting future events (Benbouzid, 2018). This self-exciting point process suggests that more events in the past increase the likelihood of future events, indicating that future occurrences depend on historical interactions (ibid). Algorithms like this which examine historical data, identify trends, and forecast future events have been used for hiring decisions, banking strategies on past loan and insurance decisions, shopping behaviours that influence future prices, and admissions acceptances as well (Wichter et al, 2018). PredPol, a predictive policing software developed by the police department in California, United States and used by law enforcement agencies worldwide uses crime hotspots to allocate resources effectively. The dangers of linking demographically related crime patterns to social factors such as gender, race, or class has resulted in greater oppression of certain social strata (ibid). Due to the algorithm's focus on demographic-related crime trends, historically over-policed neighbourhoods could lead to further subjugation of marginalised communities (Mohler et al, 2018).

Since LLMs are intended to cover a wide range of topics and groups, it is incredibly complicated to create anything approaching an adequately representative evaluation due to their loose link to an area of application (Bergman et al, 2023). Interpretations in machine learning can vary widely, being typical, characteristic, proportionate, or generalizable (Chasalow & Levy, 2021). The term "representativeness" has normative connotations in accountability and justice that needs careful consideration (ibid). Words central to data and algorithms, like "fair" and "biased," often resist precise definitions, posing challenges. As Lipton and Steinhardt caution, "suitcase words" combine multiple meanings, necessitating clear unpacking to avoid miscommunication (ibid).

Due to the availability of vast amounts of high-quality English data and resources, there is a disparity in the model performance, with non-English languages receiving less attention and resulting in inferior language capabilities (Schneider, 2022). This leads to unidimensional perspectives and characteristics that can shape the models to be significantly more effective in English-centric tasks (ibid). This results in a significant representational harm where these demographic groups are less accurately understood and represented by these technologies (ibid). The dominance of English in AI creates the belief that English speakers are superior,

making it seem normal to only speak one language and marginalising those who are bilingual or multilingual (Siegel, 2006). This idea is shaped by the belief that the language of dominant groups, like the upper middle class and native English speakers, is better (ibid). This cultural bias discourages the use and development of other languages, affecting how diverse linguistic practices are accepted and integrated (ibid).

To mitigate these harms, it is crucial for AI creators to be deliberate about how users perceive the identity behind the AI's language. Assigning cultural identities to pedagogical robots or agents is common, but unembodied language sources are often assumed to lack identity. This assumption of objectivity fails under scrutiny (Gebru, 2019). AI creators must consider all aspects of identity encoded in the language to prevent reinforcing biases and ensure fair and inclusive AI deployment. Having both an egocentric and allocentric view is crucial for decision-making and establishing a consistent identity (Iachini et al, 2023). An egocentric view focuses on the AI's perspective and immediate surroundings, enabling personalised and context-aware responses (ibid). An allocentric view, on the other hand, considers the broader environment and the position of the AI within it, fostering a more holistic understanding (ibid). Balancing these perspectives helps AI systems interact effectively within diverse contexts and maintain a stable, coherent identity across different scenarios (ibid).

E. Toxicity

LLMs have been groundbreaking in their ability to understand and generate human-like text, impacting sectors such as healthcare, education, and customer service profoundly. However, the adoption of these models has created significant risks, particularly concerning the safety and reliability of their outputs. Users often depend on these systems for crucial information, making it imperative to understand and mitigate any potential harms that may arise from their use.

Toxic generations include profanities, identity assaults, sleights, insults, threats, sexually explicit content, demeaning/belittling language, language that incites violence, and aggressive/harmful words directed towards an individual or group (Weindinger et al, 2021). Toxicity does not occur in isolation but is often influenced by the characteristics attributed to the model. A detailed analysis of over half a million responses from ChatGPT reveals a concerning trend: the model's output can dramatically change in tone and content based on the persona it adopts (Deshpande et al, 2023). For example, configuring ChatGPT to reflect the persona of a famous boxer like

Muhammad Ali was found to result in a sixfold increase in toxic responses (ibid). These toxic outputs are not just a theoretical concern but a real issue that can lead to the spread of harmful stereotypes and offensive dialogues. This variation indicates a deep-seated issue within the training of the model itself, where certain traits or identities are more likely to trigger negative and harmful content (ibid). Moreover, the problem extends beyond individual personas. The model exhibits ingrained biases that lead it to target specific groups disproportionately, exacerbating issues of discrimination and bias. This toxicity is particularly alarming because it suggests that LLMs can inadvertently or maliciously be used to propagate harmful content. It has also been demonstrated that toxicity mitigation strategies reinforce discriminatory biases, as toxicity detection instruments frequently mistakenly label statements made by historically marginalised groups as harmful (Weindiner et al., 2021).

The tendency to memorise content, combined with the fact that they are trained on publicly available internet data, makes LLMs prone to generating harmful material (Weindinger et al, 2021). The rise of highly capable, multilingual LLMs necessitates the rapid and effective scaling of toxic language detection across the numerous languages these models support (De Wynter et al, 2023). This problem is particularly acute in multilingual contexts where these models are especially likely to produce unsafe content, notably in languages with fewer resources (Jain et al, 2024). Although there have been efforts to fine-tune these models with multilingual data, the creation of datasets that adequately capture cultural nuances remains a persistent challenge (ibid)

F. Policies Surrounding AI

Governments all around the world are working to develop and implement laws, policies, and regulations that will safeguard citizens and stakeholders from the potential risks posed by AI. The field of regulation of generative AI is still in its infancy and is mainly unregulated (Pamment, 2020). In the EU, the General Data Protection Regulation (GDPR) has been the main regulation providing a framework for data protection, consent and privacy, and mandating transparency and accountability. The Digital Services Act (DSA) is a legislative framework that provides the EU with regulatory authority over digital platforms. It addresses online disinformation by imposing obligations on very large online platforms (VLOPs) and very large online search engines (VLOSEs) to conduct risk assessments and implement mitigation strategies against the spread of false information (European Parliament, 2022). These have been essential for combating the spread of false information and preventing the misuse of personal data (Voigt &

von dem Bussche, 2017). The EU also introduced the AI Liability Directive (AILD) and the Product Liability Directive (PLD), designed to provide consistent principles for liability in cases involving harm produced by AI technologies (Duffourc & Gerke, 2023). It aims to adapt existing liability frameworks to the unique challenges posed by AI technologies caused by complexity of AI systems, and to hold manufacturers accountable for defects in AI products that cause harm, respectively (ibid). However, they fail to fully address all scenarios of AI-related harm and achieve uniformity in liability for the damages caused (ibid). In 2024, the EU passed the AI Act, a landmark judgement that is the first to propose comprehensive laws for AI governance. The following section will discuss 1.) the Code of Conduct, which acts as a foundational ethical guideline for AI developers and 2.) the EU AI Act, both of which provide an understanding of the core existing guidelines for AI development in Europe. The Code of Conduct depicts ethical development practices from the ground up, while the EU AI Act offers a structured legal framework to ensure these technologies are implemented safely and responsibly.

1. CODES OF CONDUCT (CoC)

CoCs provides a set of guidelines and principles intended to ensure that engineers follow ethical standards in the development of LLMs. The code is essential to uphold basic ethical tenets but insufficient because it provides a value based approach but lacks normative orientation (Gogoll et al, 2021). A study found “accountability”, “privacy”, and “fairness” are mentioned in approximately 80% of all guidelines, suggesting these principles are unanimously considered essential for creating ethically sound AI systems (Hagendorf, 2020). Another study corroborated this by analysing 84 documents related to AI ethics, identifying eleven primary principles: transparency, justice/fairness, non-maleficence, responsibility, privacy, beneficence, freedom/autonomy, trust, sustainability, dignity, and solidarity, and found "transparency" to be the most frequently mentioned principle, appearing in 87% of the documents (Jobin et al., 2019).

CoCs indicate a shared understanding of the foundational values necessary for the ethical development and deployment of AI technologies but often fail to give sufficient guidance on how to balance these values in practice. This lack of concretization and contextualization leaves software engineers to navigate the complexities of adhering to multiple, sometimes conflicting, values on their own (ibid). Ethical principles such as privacy, transparency, autonomy, and safety are frequently listed in CoCs, but the inherent tensions between these values create

significant challenges during implementation (Kearns & Roth, 2019, Gogoll et al, 2021). For instance, achieving both privacy and transparency simultaneously is often impractical, as illustrated by the trade-offs depicted in Figure 2.

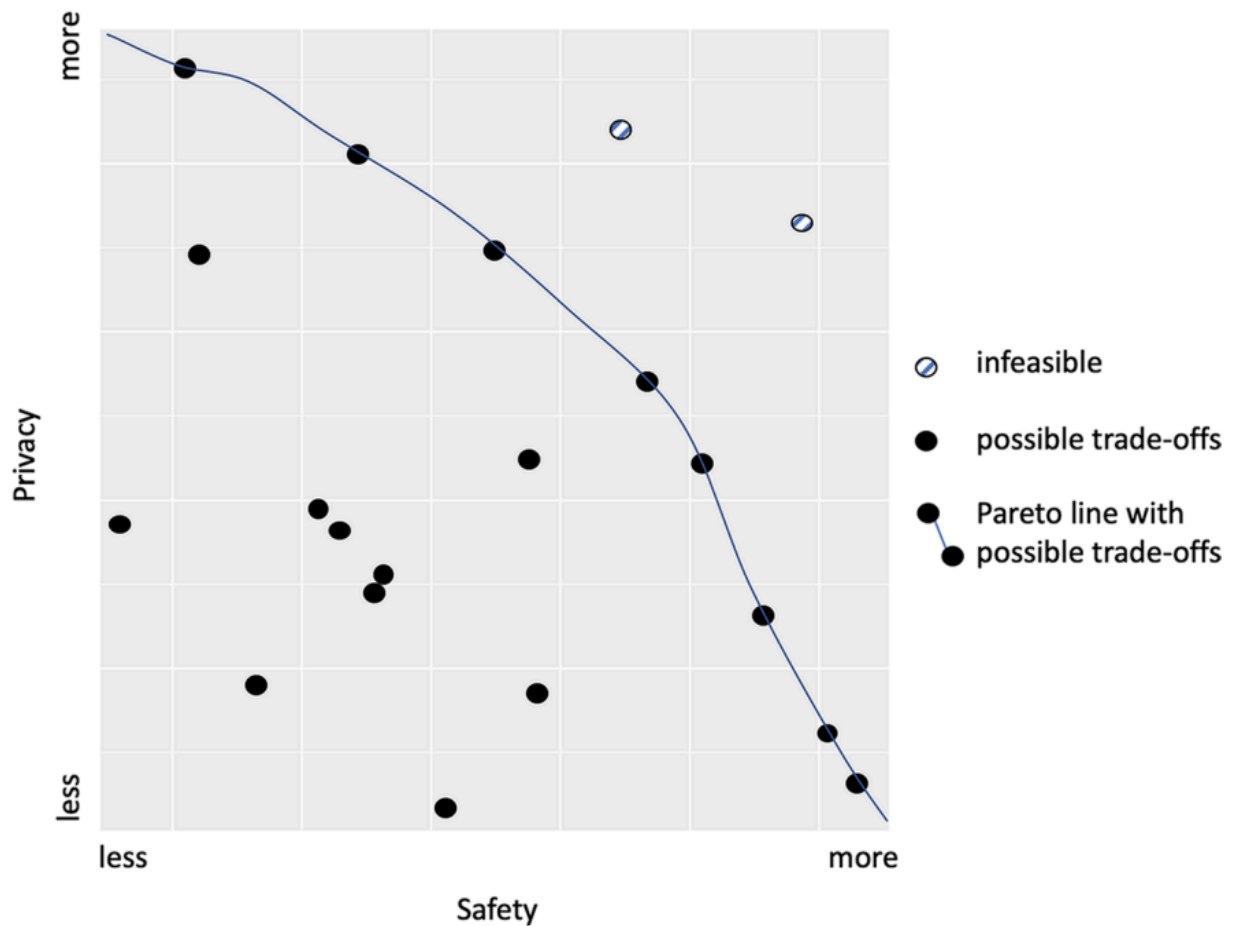


Figure 1. (Gogoll et al, 2021)

The diagram highlights the concept of Pareto optimality, where any improvement in one value necessitates a compromise in another once the efficiency frontier is reached (Gogoll et al, 2021). This trade-off is particularly evident in scenarios such as geo-tracking applications, where the balance between privacy and transparency varies greatly depending on the use case (ibid). On the other hand, an app used to monitor professional athletes for anti-doping purposes might justifiably prioritize transparency over privacy, whereas a navigation app for general public use would likely prioritize privacy to protect users from constant monitoring (ibid).

This variability underscores the need for contextual ethical deliberation and decision-making, which CoCs fail to adequately address. As a result, developers are left without the necessary tools to make informed ethical decisions, and often resort to cherry-picking, indifference, depending too much on instinct, and ex-post orientation; when values do not align (ibid). Critical decisions are often made in an ad-hoc manner, lacking consistency and robustness leading to potential ethical lapses and public distrust (ibid).

2. *AI ACT, 2024*

The act lays down comprehensive rules in its 459 page document, outlining rules for safety, ethical considerations, and fundamental rights, according to the level of risk the AI system poses. The act classifies models in four main risk groups; unacceptable, high, limited and minimal risk (European Commission, 2024). The provides legal backing to enforce non-discrimination laws and supporting the technical fairness discourse, ensuring that ethical principles are not just theoretical but are implemented and regulated (ibid).

The AI Act targets AI systems that pose an **unacceptable risk** to values like human dignity, democracy, and fundamental rights. For instance, AI that manipulates behaviour causing harm or exploits vulnerable groups, like children or the elderly, is banned (Almada and Petit, 2023). The majority of the AI Act's provisions target models operating at 10^{25} FLOPS, a threshold that currently includes only two prominent models: GPT-4 by OpenAI and Bard by Google (Hacker, 2023). This categorization means that popular models like GPT-3, Bard, Copilot, and Mistral are not subject to rigorous regulations and without stringent guardrails potentially expose users to higher risks (ibid). **The high risk category** covers AI systems posing threat to health, safety, and fundamental rights. These systems can only be marketed if they meet strict legal requirements, including technical specifications and the creation of mechanisms to monitor risks (ibid). The high-risk classification is expected to cover 5% to 15% of AI systems in the EU, aiming to harmonise rules across member states (Almada and Petit, 2023). All other AI systems fall under **limited and minimal risk** where it does not impose specific rules for these systems but establishes guiding principles (ibid). These lower-risk AI systems are regulated by other EU and national laws, such as the General Product Safety Regulation (ibid). Additionally, Member States can adopt AI-specific rules in areas not covered by EU-wide regulations (ibid). This risk based approach is similar to other EU digital regulations like the General Data Protection Regulation (GDPR), the Digital Services Act (DSA), and the Digital Markets Act (DMA) (ibid). The Digital Services Act does address more concerns in a decentralised way so it applies to all

models, including content moderation rules for illegal content (ibid). However, there is an urgent need to mandate safety regulations for models posing all or no types of risk.

Furthermore, the AI Act does not provide a clear definition of misinformation or specify the types of content that pose the greatest risks to democratic processes and civic discourse. To better protect against misinformation, the policy could include a comprehensive definition of misinformation and specify the criteria for identifying problematic content, such as false information, misleading narratives, or malicious propaganda (Wardle & Derakhshan, 2017). Additionally, the policy should include a systematic framework for assessing the risks associated with misinformation and evaluating the effectiveness of mitigation measures. This framework could establish a risk assessment methodology that considers factors such as the potential impact of misinformation on public opinion, electoral processes, and social cohesion, guiding regulators and stakeholders in identifying high-risk scenarios and implementing targeted interventions (Marwick & Lewis, 2017).

While the act emphasises the importance of transparency and accountability in AI systems, it does not specify how these principles will be operationalized in the context of misinformation. To enhance transparency and accountability, the act could require AI providers and deployers to disclose information about their content moderation practices, including the criteria used to identify and label misinformation, the effectiveness of detection algorithms, and the processes for reviewing and appealing content decisions (Pasquale, 2015). Furthermore, mechanisms should be established for independent oversight and accountability, such as third-party audits or regulatory monitoring, to ensure compliance with transparency requirements and safeguard against the misuse of AI systems for spreading misinformation (Binns, 2018). There is also a need for collaborative governance models that involve government agencies, technology companies, civil society organisations, and academic researchers in joint efforts to address misinformation. This could include establishing platforms for information sharing, coordinating research initiatives, and facilitating multi-stakeholder dialogues to develop coordinated strategies for combating misinformation (Helberger et al, 2018).

One of the significant challenges in identifying automated discrimination in AI systems stems from their non-decomposable nature (Wachter et al, 2020). Complex modern machine learning or AI systems cannot be easily broken down into isolated components (ibid). In other contexts apart from AI, the judiciary can help regulation by decomposing large systems into individual policies, evaluating each for compliance with non-discrimination laws (Barocas, Hardt, &

Narayanan, 2019). However, AI systems often lack this decomposability, complicating the understanding of discrimination they may cause. Proving automated discrimination becomes difficult without accessible and relevant statistical evidence. Claimants often lack information about a system's optimization conditions or decision rules, leaving them unaware of the exact rule that led to perceived disparity (ibid). This absence of transparency and detailed information limits the ability to demonstrate and prove discrimination within complex AI systems.

Furthermore, the AI act is designed as a product safety framework suggesting that AI systems have specific, well-defined purposes, which do not align with the nature of general-purpose AI systems that can be used in various contexts (Almada and Petit, 2023). This safety-focused approach may lead regulators to frame AI issues in terms of safety risks, which is not suitable for addressing fundamental rights concerns that the AI Act aims to protect (ibid). The regulatory structure might rely too heavily on existing product safety institutions, failing to recognize the need for developing new capabilities specifically for AI to ensure the protection of fundamental rights (ibid).

One of the most forward thinking suggestions of the AI Act is that of 'Regulatory sandboxes' which serve as experimental frameworks within which AI technologies can be tested and evaluated under regulatory oversight but with reduced constraints (OECD, 2023). This approach allows developers to explore innovative AI applications while still ensuring that these products meet safety and ethical standards before wider deployment. The purpose of such sandboxes is dual: they facilitate technological innovation by reducing initial regulatory barriers and simultaneously protect public interests by ensuring that new AI applications do not compromise consumer safety or fundamental rights (ibid). The impact of these sandboxes is significant as they accelerate the development of AI by allowing quicker adaptation to a fast-evolving field. This informed regulation helps to minimize the risk of fragmentation across different jurisdictions, which can impede the ongoing 'AI race' by creating inconsistent standards (ibid).

G. Social Impact

Throughout history, there has been a persistent concern that technological advancements would displace human workers and cause irreversible harm to the labour market. Initially it was mechanisation, computing, robotics and now it is AI that has captured the imagination of people worldwide. John Maynard Keynes described "technological unemployment" as job loss caused by the rapid pace of technological progress outstripping the creation of new labour opportunities

(Furman and Seamans, 2019). Despite these fears, automation has often led to short-term job displacement but ultimately resulted in the creation of complementary jobs in the long term (Autor, 2015). Historical evidence shows that automation has had varying impacts depending on the occupation. Pushing the frontiers in AI could similarly enhance productivity, as demonstrated by the positive impacts of past technological innovations (ibid).

The primary challenges that hinder AI regulation include the opaque nature of AI systems, the necessity of respecting human agency, and the perpetuation of social inequalities through AI (Ghosh and Lakshmi, 2023). Generative AI systems often operate as 'black boxes,' making it difficult to understand and scrutinise their decision-making processes. This opacity can be exploited by organisations to evade accountability, as seen in cases like Apple's discriminatory credit card system and Lufthansa's anti-monopoly issues (ibid). To counter these challenges, there is major work underway on the development of transparent and explainable AI systems, which can provide on-demand insights into their operations (Barredo Arrieta et al., 2019; Miller, 2019). However, there are significant technical and legal challenges in unpacking these black boxes, due to its complexity and privacy issues. A more achievable target in the near future should be laying out explanations to help data subjects act and providing the scope and content of explanations related to the action or goal they're dealing with (Wachter et al, 2018). This approach aligns with the principles of Human-Centred AI (HCAI), which aims to make AI safer and more sustainable by focusing on human-centric design, development, and deployment (Wao, 2023). By incorporating broader social structures, policies, ethics, and legal frameworks, a sociotechnical approach ensures that AI systems are developed and implemented with a deep understanding of their potential societal impacts.

This predominant focus on technical fixes for ethical challenges like accountability, explainability, and fairness (Sartori, 2022). These aspects are frequently highlighted because they can be operationalized mathematically, thus fitting within the existing technical frameworks (Mittelstadt et al., 2019). The underrepresentation of non-technical ethical considerations has caused many critical issues to go unaddressed; the hidden costs associated with e-waste, energy consumption, and the exploitative labour practices tied to data labelling and content moderation (Crawford and Joler, 2018). The omission of these factors can perpetuate systemic inequalities and fail to account for the full spectrum of AI's societal impact. Relying solely on technical metrics for fairness can blur the lines between fairness as a mathematical property and fairness as a social ideal (Green & Hu, 2018). Fairness as a mathematical property involves specific, quantifiable metrics and algorithms to ensure that decisions are unbiased according to

predefined criteria (ibid). These criteria often include statistical measures such as equal opportunity, demographic parity, and calibration (ibid). Fairness as a social ideal encompasses broader, context-dependent ethical and moral considerations. It involves understanding the historical, cultural, and social contexts that contribute to inequality and striving to achieve justice and equity beyond just statistical measures (ibid). This confusion is evident in how the field has interpreted the "impossibility results" which demonstrate that it is impossible to satisfy multiple statistical fairness criteria at the same time (Kleinberg et al., 2016; Chouldechova, 2017). Although mathematical fairness metrics is a starting point, it must be contextualised within broader social ideals. This emphasises the need for a deeper understanding of the social implications of AI decisions, achieved by engaging with stakeholders, considering the historical context of marginalised groups, and ensuring that AI systems are monitored and held accountable for their impact.

Systemic inequalities also manifest due to the gender imbalance within the AI research community, influencing the development of these guidelines. The discourse on AI ethics is largely shaped by a homogenous group, often leading to guidelines that reflect a narrow set of perspectives. This imbalance is more than a mere demographic issue; it reflects a deeper philosophical divide. Men are known to approach ethical problems through a "calculating," "rational," and "logic-oriented" ethics of justice, whereas women are more likely to interpret these issues within a framework of "empathic," "emotion-oriented" ethics of care (Giligan, 1982). This difference can result in ethical guidelines that do not fully address the needs and concerns of all affected groups, emphasising rule-based justice over relational and contextual ethics (Sartori, 2022). There is a need for multi-stakeholder governance in AI to create reliable AI and increase the inclusiveness and quality of decision-making (OECD et al, 2022) To keep up with these changes, governments need to think about how to upskill and reskill women in accordance with their labour markets, regulations, and skill mixtures (ibid). Furthermore, companies must consider how to apply AI to develop skill-equalising work settings for women (ibid). LLMs risk impeding social change by codifying and upholding ideals from a specific group or historical period (Weidinger et al, 2021). Once these correlations are embedded, it makes it more difficult to reclaim terminology or create inclusivity (ibid). This affects social movements that greatly rely on language as a tool to shift power and assign confidence (Bender et al., 2021).

Most guidelines provide minimal concrete technical explanations or actionable steps for implementing ethical principles (Brundage et al., 2018). This gap between ethical principles and

practical implementation makes it challenging for practitioners to translate these guidelines into effective, real-world applications. The lack of detailed technical instructions can lead to inconsistencies in how ethical principles are applied across different AI systems. These issues collectively lead to a fragmented approach to AI ethics, where guidelines fail to integrate technical solutions with broader social and ecological considerations. This fragmentation can result in the continued perpetuation of biases, discrimination, and other ethical risks, ultimately undermining the trust and effectiveness of AI systems in society. Addressing these issues requires a comprehensive approach that includes diverse perspectives, detailed technical guidance, and consideration of the broader social and ecological contexts in which AI operates.

Cory Doctorow, a futurist, comments on the debate between 'Effective Altruists'; those that advocate for rigorous AI safety measures to ensure these advanced systems become beneficial partners rather than threats and "Effective Accelerationism"; those that push for rapid AI development with minimal safety protocols, believing this will drive technological and economic progress (Doctorow, 2023). He argues that beyond algorithmic ethics, the critical focus should be on the ethical foresight in deploying these technologies; in ensuring fairness, accountability, transparency, and interpretability in AI systems (ibid). Moreover, the real risk often lies in the application by those in power, emphasising the need for robust governance frameworks. The difference between an image recognition system being utilised for humanitarian rescue or military aggression, highlights the importance of ethical deployment decisions made by policymakers, not merely the AI itself (ibid). For this reason, the integration of sociotechnical frameworks is pivotal in addressing the ethical risks associated with generative AI, particularly in mitigating biases related to misinformation, representation, and toxicity. Technical solutions alone are insufficient to address the complex social issues that arise when AI systems are deployed (Selbst et al., 2019). Traditional fairness-aware machine learning approaches often fall into abstraction traps by neglecting the social context in which these systems operate (ibid). This oversight leads to technical interventions that may be ineffective or even harmful when applied to real-world scenarios (ibid).

Incorporating the sociotechnical perspective, it becomes evident that addressing ethical risks in AI involves more than just technical solutions. A sociotechnical approach integrates broader social structures, policy, ethics, stakeholders, legal considerations, cultural considerations, roles, and goals into the design and deployment of AI systems (ibid). When researchers focus solely on the technical aspects and disregard the social environment, they make an "abstraction error" (ibid). This means they incorrectly separate the technical system from the social systems it

affects, leading to incomplete and potentially flawed understandings and implementations of fairness. Without a sociotechnical perspective that considers both the technology and its social implications, it's impossible to fully grasp or achieve true fairness (ibid). This gives rise to the importance of studying AI through the lens of abstraction; simplifying complex, domain-specific problems by removing detailed context so that machine learning tools can be universally applied (Selbst et al., 2019). This practice is fundamental to the versatility of ML, but it also introduces significant ethical challenges when the social context is inadequately considered (ibid). These traps illustrate the pitfalls of relying on abstraction in AI systems and highlight the necessity of incorporating nuanced understanding and ethical considerations into design and deployment (ibid).

Solutionism: Involves the inappropriate application of technical solutions to social problems without adequately understanding the social context and its complexities (ibid). For example, the mathematical definition of fairness might focus on statistical parity or equalised odds, but these definitions can diverge significantly from social understandings of fairness, which encompass broader issues of justice and equity. This divergence can lead to insufficient efforts in making AI systems truly fair. To mitigate this, it is essential to conduct thorough legal and social research before implementing technical solutions, ensuring that they are contextually appropriate and exploring non-technical alternatives when necessary (Morozov, 2013). In the context of generative AI, this might involve stakeholder consultations to understand the societal impact of AI-generated content.

Ripple Effect Trap: The ripple effect occurs when the introduction of a technical solution leads to predictable changes in the social context, potentially exacerbating existing problems (Selbst et al., 2019). A notable example is the COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) tool used to forecast recidivism rates among inmates (Skeem & Lowenkamp, 2016). It was intended to assist in pre-trial risk assessments but ended up propagating biases related to race and gender. Research showed that black defendants were mistakenly flagged as potential re-offenders nearly twice as often as white defendants (Angwin et al., 2016). This issue extends to vulnerable communities when data analytics recommend police interventions in specific districts, often linked to crime through grid-mapping visualisations (Barabas et al., 2018). Predictive policing may reinforce existing biases and amplification of pre-existing biases that promote "evidence-based stigma," which can be detrimental to individual and community development (ibid). The manifestation of these biases can be attributed to the fact that COMPAS is viewed through competing statistical fairness metrics.

COMPAS might be considered fair by one measure (calibration), but the algorithm's accuracy doesn't account for historical and cultural contexts, such as the long history of discrimination against black people (ibid). Accurate predictions of recidivism still perpetuate the effects of systemic racism, as these predictions can lead to disproportionate imprisonment of these individuals.

Formalism Trap: Formalism involves the over-reliance on rigid, formal rules to define concepts like fairness, which are inherently procedural, contextual, and contestable (Selbst et al., 2019). AI systems must incorporate flexible, context-sensitive definitions of fairness that allow for procedural justice and adaptability to different social settings. For instance, the conflicting statistical definitions of fairness, overshadows other critical aspects like high misclassification rates and the overall justice of a tool's purpose (Green & Hu, 2018). This led to neglect of other key aspects of the tool's behaviour and performance that are relevant to its fair deployment (ibid). This fixation on statistical parity neglects broader procedural and contextual considerations, and demonstrates the need for flexible, context-sensitive definitions of fairness that involve diverse stakeholder perspectives to ensure just AI deployment (Mulligan et al., 2019).

Portability Trap: Portability is the challenge of transferring an AI solution from one context to another without appropriately adapting it to the new environment (Selbst et al., 2019). An AI tool developed for one jurisdiction might not be suitable for another due to differing social and legal landscapes. To avoid this trap, developers must ensure that the social and technical requirements of the target context are thoroughly understood and integrated into the AI system (ibid).

Framing Trap: Framing refers to the narrow definition of the problem and the exclusion of relevant data and social actors in the AI system's design. This can lead to solutions that do not adequately address the actual issues or needs of the community. To counteract this, the problem must be framed heterogeneously, including all relevant stakeholders and data, to ensure that the AI system addresses the localised question of fairness comprehensively (Benford & Snow, 2000).

To effectively tackle these traps, it is essential to thoroughly document and communicate the limitations and potential impacts of any technical solution. This includes considering alternative policies and solutions that might better address the underlying issues without relying solely on

technological fixes. By doing so, we can ensure that AI systems are not only technically robust but also socially and ethically aligned with the communities they serve.

METHODOLOGY

To conduct a comprehensive evaluation of current methods in AI ethics, a systematic study of over 170 academic papers was performed.

The Sociotechnical Repository served as a primary source for gathering papers. Additionally, papers not available in the repository but relevant to the study were included, ensuring a holistic view of the current landscape. Papers were chosen based on their relevance to AI ethics, the methodologies they employed, the nature of their findings related to ethical risks, and any proposed evaluation strategies. Papers were selected by conducting keyword searches on Google and the Sociotechnical Repository. Keywords included terms like "AI ethics evaluation," "Sociotechnical harms," "Misinformation," "Representational errors", and "Toxicity".

A systematic analysis was conducted to identify patterns in the evaluation methodologies used across different studies. This involved grouping papers based on their approach to evaluating ethical risks and the outcomes they reported. By analysing patterns in the evaluations, a taxonomy was developed to categorise different dimensions of AI ethics evaluations. This taxonomy served as a framework for assessing the strengths and weaknesses of various evaluation strategies.

An extensive qualitative analysis was performed to explore the methodologies and findings of each paper. This approach enabled a thorough understanding of the underlying patterns and relationships, which are often missed by quantitative methods. Each evaluation method was scrutinised for limitations, providing insights into the areas that require improvement or more nuanced approaches. Based on the findings and the identified limitations, a comprehensive set of evaluation guidelines was developed. These guidelines are designed to be adaptable and to evolve with new research and emerging ethical standards.

EVALUATION STRATEGIES

A. MISINFORMATION

Detecting Misinformation in LLMs

Ensuring the factual accuracy and integrity of the generated content has become a paramount part of LLM development. Based on my analysis, there are three primary methodologies to detect misinformation: **Comparison to Ground Truth**, **Linguistic and Semantic Analysis** and **Safety and Responsibility Evaluations**. Detecting misinformation in AI-generated content involves verifying the factual accuracy, examining the linguistic structure of sentences and ensuring that there is responsible deployment considering context and comprehensiveness. By comparing generated content to a verified ground truth, we can assess the factuality and validity of the information. Additionally, analysing the linguistic and semantic structure helps identify inconsistencies and inaccuracies that might not be immediately evident through simple fact-checking. These approaches are essential for maintaining the credibility and reliability of AI systems.

Comparison to Ground Truth

- **Converting a Fact Corpus into a Benchmark to Assess Factuality:** Multiple researchers evaluated the factual accuracy of LLMs by creating benchmarks from a factual corpus. One such method involves selecting a prefix text from the corpus and generating multiple sentence completions, only one of which is factually correct. The model's performance is measured by how often it identifies the correct completion. This process helps in determining how well the model can distinguish between factual and non-factual information. The FACTOR framework ensures that non-factual completions are grammatically correct and similar to factual ones, making it harder for the model to rely on superficial cues.
- **Hallucination Detection Benchmarks:** Hallucination detection benchmarks are designed to test the performance of classifiers in identifying incorrect or fabricated content generated by LLMs. These benchmarks provide a systematic way to evaluate how often models generate hallucinated facts and measure their ability to distinguish between accurate and inaccurate information. This method helps in understanding the

propensity of LMs to produce factually incorrect statements and guides improvements in model training to reduce such occurrences.

Linguistic and Semantic Analysis

- **Model-Based Metrics (Entity Recognition and Relation Classifier):** Linguistic and semantic analysis involves using both model-based and model-free metrics to evaluate the factual accuracy of text. Model-based metrics, such as entity recognition and relation classifiers, break down the text into entities and relationships and compare these to the expected entities and relationships in the factual data. This approach uses another model to evaluate the generated content, focusing on named entities and the correctness of their relationships.
- **Model-free metrics, like Open Information Extraction (Open IE) and ROUGE,** do not rely on another model for evaluation. Instead, they use predefined metrics to assess the quality of summaries by measuring overlap with n-grams, word sequences, and word pairs in the reference text. This method is straightforward and interpretable, making it easier to apply across different models without additional training.
- **Fact-Checking Tools:** RARR (Retrieval-Augmented Rewriting) for instance, revises AI-generated text by gathering relevant evidence and ensuring that the content aligns with this evidence while maintaining its style and structure. This tool allows for seamless integration of accurate information, effectively replacing the original text with a fact-checked version. RARR exemplifies how retrieval-augmented models can enhance the factuality of generated content without altering the fundamental capabilities of LLMs.

Safety and Responsibility Evaluations

The safety and responsible evaluation of AI-generated content is crucial in preventing misinformation, especially in sensitive areas such as healthcare. Misinformation is not just about the factual accuracy of content but also how it is disseminated and contextualised. Ensuring responsible deployment involves providing comprehensive information and appropriate caveats to avoid misleading interpretations.

A notable example of the risks associated with inadequate safety and responsible evaluations is seen in the dissemination of information related to Alzheimer's disease. A study by Huang et al. (2023) revealed that while the content generated by AI on Alzheimer's was not factually incorrect, it omitted key information and failed to follow ethical guidelines essential for medical

advice. This omission of critical details and lack of context can lead to confusion among care providers, potentially resulting in dangerous situations (Huang et al., 2023). Additionally, models are known to exhibit sycophantic tendencies, often aligning with user ideologies and beliefs. This behaviour can be particularly problematic when dispensing medical advice, as it may reinforce incorrect or biased views rather than providing objective and comprehensive information (Huang et al., 2023).

Supporting these findings, Spalek et al. (2023) evaluated AI-generated medical information based on metrics like communication of guidelines, readability, and quality of advice. The study found that while AI-generated content might appear satisfactory at face value, it often does not meet the standards of professional medical advice. The AI outputs lacked the depth and contextual understanding provided by human doctors, highlighting the need for strict human oversight and engineered prompting to ensure safe deployment (Spalek et al., 2023).

To address this, it is recommended to implement misinformation detection algorithms and design prompt interventions in a risk triage manner (ibid). AI systems could provide warnings or public service announcements about potential misinformation, directing users to verification resources like fact-checking websites and reputable news organizations. Current off-the-shelf large language models (LLMs) acknowledge the possibility of generating misleading outputs through disclaimers. For instance, OpenAI warns that their models "sometimes writes plausible-sounding but incorrect or nonsensical answers," and Galactica notes "generated text may appear very authentic and highly-confident, but might be subtly wrong in important ways" (Zhou et al, 2023). However, these disclaimers are insufficient, as they could shift the responsibility of verifying information to users, creating a false sense of user knowledge and consent regarding AI capabilities and limitations (ibid). For high-risk misinformation content, such as the false claim that "drinking bleach is a cure for COVID-19," measures like blocklists can limit such content during the generation stage, or the model can offer direct corrections with explanations and links to credible sources (ibid).

Detecting misinformation in AI-generated content requires a multi-faceted approach. By comparing generated text to verified ground truth and employing linguistic and semantic analysis, we can significantly enhance the factual accuracy and reliability of language models. These methodologies not only help in identifying and mitigating misinformation but also contribute to the development of more robust and trustworthy AI systems.

Evaluating Misinformation in LLMs

Evaluating misinformation involves verifying the factuality of the content, ensuring its accuracy, and examining its linguistic structure. The primary methods used to evaluate misinformation in LMs include benchmarks for factuality assessment, model-based evaluations using classifiers, and model-free evaluations using deterministic metrics. The rise of advanced LLMs have revolutionised text generation and information retrieval. However, the potential for these models to spread misinformation necessitates robust evaluation methods.

Benchmarks for Factuality Assessment

- **WIKIFACTOR and Expert-Curated Datasets** : WIKIFACTOR and similar benchmarks assess the factuality of generated content by comparing it to a curated set of facts. These benchmarks typically involve multiple-choice questions where models must select the correct factual completion from a set of options. For instance, the FACTOR framework generates benchmarks by selecting a prefix text and generating multiple completions, only one of which is factually correct. Models are evaluated based on their ability to identify the correct completion, with non-factual options crafted to be grammatically correct and similar in style to increase difficulty. Following a structure similar to this, provides a clear measure of factual accuracy. It is useful for evaluating a model's ability to distinguish between factual and non-factual content. However it is difficult to create such high-quality benchmarks as they are resource-intensive, and as a result may not cover the full range of potential misinformation scenarios.
- **POLITIFACT and Domain-Specific Datasets**: POLITIFACT and other domain-specific datasets are used for political and general factuality testing. These datasets include verified information on various topics, allowing for precise evaluation of model outputs. It is efficient because it leverages existing fact-checked information. Additionally it allows for specific evaluations in politically sensitive contexts. However, it is limited to the domains and topics covered by the dataset and it may not adapt well to rapidly evolving misinformation topics.

Model-Based Evaluations

- **Relation Classifiers and Named Entity Recognition**: Model-based evaluations involve using classifiers to assess the relationships and entities in generated text. Named entity recognition (NER) identifies entities in the text, and relation classifiers evaluate the

correctness of relationships between these entities. It provides detailed insights into the factual structure of text. The key lies in detecting nuanced factual errors involving entity relationships. Conversely, the complexity in implementation could lead to potential for cascading errors from the evaluation model. This evaluation requires additional models and training data for evaluation.

Model-Free Evaluations

- **ROUGE Metrics for Summarization Quality:** ROUGE metrics provide a deterministic way to evaluate the quality of text by measuring the overlap with reference summaries. This method assesses the generated text's quality based on n-grams, word sequences, and word pairs. It is a simple and interpretable metric for evaluating text quality. It also does not require additional models for evaluation. The drawback of this method is that it focuses on text similarity rather than factual correctness. This is a flaw as it may not accurately capture the factual accuracy of content.

Challenges in Detecting Misinformation

The complexity of language, domain-specific variations, ethical dissemination, and the inherent limitations in evaluation metrics all contribute to the difficulty of detecting misinformation. Each challenge presents unique difficulties and requires distinct strategies for effective detection. Understanding these challenges is essential for developing more effective detection tools and methods.

Language Complexity

- **Authoritative and Less Sentimental Language:** AI-generated misinformation often mimics authoritative and neutral tones, making it challenging to detect. Such language can obscure falsehoods, as it appears credible and trustworthy. A study revealed that AI-generated misinformation using authoritative language significantly complicates detection efforts, as traditional fact-checking methods may overlook nuanced falsehoods embedded in otherwise credible-seeming text (Zhou et al, 2023).

One of the significant challenges is analysing the linguistic structure of both human and AI-generated misinformation. To analyse the linguistic structure of human misinformation and AI generated misinformation, three different types of patterns were tested in news stories, social

media posts, and human misinformation (ibid). It was carried out for validating the context and identifying fake information by comparing it with known accurate versions. However, the structures of misinformation vary widely across different domains, such as medical versus political misinformation, making it difficult to establish a one-size-fits-all detection method (ibid). AI-generated misinformation often tries to enhance credibility by citing sources and following patterns that appear more believable than human-generated misinformation. This complexity requires domain-specific analysis, which may still fall short due to the diverse and evolving nature of misinformation structures. Consequently, while linguistic analysis can help detect misinformation without always needing a reference context, the variability and sophistication of misinformation present significant hurdles in effective detection (ibid).

This language complexity can generate reliable sounding data even when trained on a misinformation dataset. Its creation of “hypothetical language and long dependencies” make it difficult for detection models to capture the context and make correct predictions (ibid).

Domain-Specific Variability

Misinformation structures vary significantly across different domains, such as politics, healthcare, and science. This variability necessitates tailored detection approaches for each domain, as a one-size-fits-all method is insufficient (ibid). This is not always feasible as it requires extensive domain-specific knowledge and datasets, thereby increasing the complexity and resource requirements for developing detection methods.

In high-risk fields like medicine, ensuring that AI-generated content adheres to ethical guidelines and medical standards is crucial. Misinformation is not just the factuality of content but also the way in which it is disseminated. A study revealed information related to Alzheimers was not factually incorrect but it omitted key information, and did not follow ethical considerations and responsible guidelines that should accompany medical advice (Huang et al, 2023). By omitting key information, and losing context, the information has a risk of making things confusing for care providers, leading to a dangerous situation (ibid). Models also tend to be sycophantic, and they have been known to agree with user ideologies and beliefs, which can pose a problem when giving out medical advice (ibid).

Another study corroborated this by judging sources based on a number of metrics like communication of guidelines, readability grade, quality of advice provided etc and found that they are not able to provide proper guidelines (Spalek et al, 2023). Although the face validity

seemed satisfactory, it was not comparable to medical opinions given by doctors (ibid). All the AI generated outcomes need strict human oversight and engineered prompting and editing to be safe to deploy (ibid).

A significant drawback is that the quality of output depends on the prompt and the materials underlying the response. This highlights privacy concerns, as to get specific or accurate responses, one has to share personal medical information (Seyyed- Kalantari, 2021).

Evaluation Limitations

- **Generalizability of Evaluation Metrics:** Creating generalizable evaluation metrics for misinformation detection is challenging due to the subjective nature of biases and the diverse types of misinformation. Existing metrics often fail to capture the full spectrum of misinformation scenarios. A study assessed the limitations of current hallucination detection benchmarks, noting that these metrics often struggle to generalize across different types of misinformation and biases inherent in the training data (Liu et al, 2022).

B. REPRESENTATION

Understanding the complexities of socio-technical systems is needed for addressing the risks and harms associated with AI as these systems' behaviour results from interactions between networks of human actors and technical subsystems. There are currently multiple taxonomies created to categorise AI risks and harms. For instance, harms can be categorised into five themes: **Representational harms**; when AI systems reinforce the subordination of social groups, **Allocative harms**; when opportunities, resources, or information are withheld from marginalised groups, **Interpersonal harms**; those that **affect** relationships between people and communities, **Quality-of-Service Harms**; when systems fail disproportionately for certain groups, **Social System Harms**; those that adversely affect society at large (Shelby et al, 2020). These taxonomies were expanded on to pin-point six high-level risk areas for large language models: representation and toxicity harms, misinformation harms, information and safety harms, malicious use, human autonomy and integrity harms, and socio-economic and environmental harms (Weidinger et al, 2022). Another way to approach harms is to consider AI privacy risks, categorising harms from incidents into data collection, data processing, data dissemination, and invasion risks (Leet et al., 2021). AI failures can be looked at from the viewpoint of **errors of**

commission; design flaws and **errors of omission**; implementation flaws (Chanda and Banerjee, 2023). Work has been done to develop a taxonomy for AI system failure analysis, combining system goals, implementation methods, and technical failure causes (Pittaras and McGregor, 2022). This was further refined for personalised large language models, adding risks such as user effort and volunteer labour, essentialism and profiling, and increased anthropomorphism (Kirk et al., 2023).

This sample of definitions brings to light the significance of not only the specific harms inflicted by AI but also the socio-technical network from which they arise; who is harmed, who is accountable, and how these roles influence the manifestation of specific harms (Hutiri, 2024). Another study categorised societal-scale risks from AI into several high-level categories, including risks related to economic stability, political stability, environmental sustainability, public safety, and fundamental rights (Critch and Russell, 2023). These harms (as depicted in the figure) are related to Diffusion of Responsibility, “Bigger than expected”, “Worse than expected”, “Willful indifference”. “Criminal Weaponization” and “State Weaponization” (ibid). Similar to this, the CSET AI Harm Framework segregates various harm types into physical health or safety, financial loss, infrastructure damage, environmental impact, violation of human rights, and other tangible and intangible harms (Hoffmann et al., 2023).

Figure 2: An exhaustive decision tree for classifying societal-scale harms from AI technology

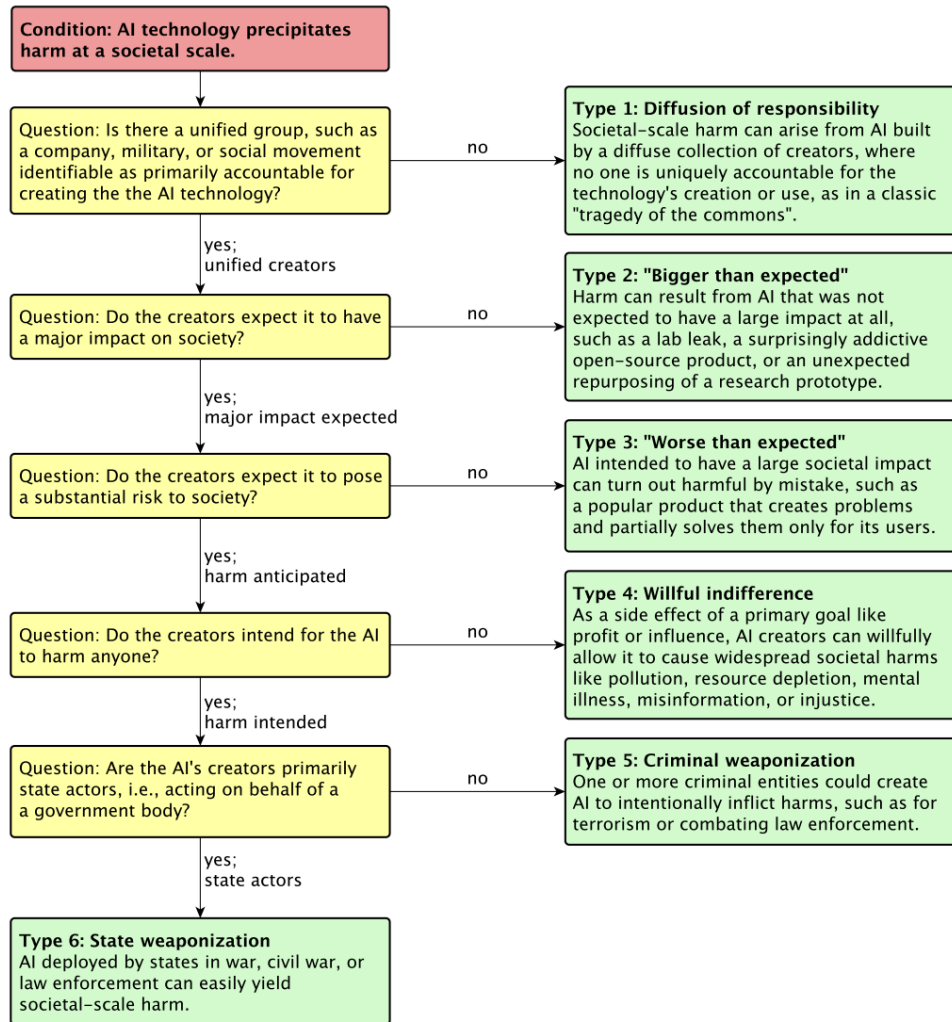


Figure 2. (Hoffman et al, 2023).

To understand and evaluate the Sociotechnical harms, each of the evaluations are also classified according to the type of bias and the axis of bias by proposing a taxonomy that can help guide future evaluation efforts.

Bias Taxonomy

Classifying evaluations based on the type of bias they aim to identify can help in understanding the coverage of the biases across different axes of social groups. Additionally, it helps facilitate

a discussion on how various studies have conceptualised and operationalised these different biases. These categories are broadly defined on purpose as bias can be difficult to specify without a clear definition of the use case and the social groups involved.

1. **Stereotypical Bias:** A model can be said to display stereotypical behaviour when it generates or reaffirms existing stereotypes in the real world.

Bianchi et al., 2023 conducts a quantitative and qualitative analysis by using neutral text prompts, some which explicitly include descriptors of human traits, occupations, and objects. They analyse the disparity between the neutral intention of user prompts and the biased stereotypical outputs generated by the model which involves comparing the characteristics of generated images against known societal demographics and stereotypes. Multimodal Composite Association Score (MCAS) (Mandal et al., 2023) is a composite score that measures associations between gender-specific attributes (e.g., 'male' or 'female') and various targets (e.g., occupations or objects) in both text and image formats. They use simple prompts containing different occupations to measure four scores to identify occupational biases towards gender; The Image-Image Association Score (IIAS) measures the similarity between the generated image and a typical set of images representing a gender; The Image-Text Prompt Association Score (ITPAS) measures the similarity between gendered images and the textual prompts used to generate images; The Image-Text Attributes Association Score (ITAAS) measures the similarity between a set of gender specific words and the generated images; The Text-Text Association Score (TTAS) measures the similarity between the gendered text set and the prompt. Sun et al., 2023 focuses on representational and presentational harms in text-to-image models, specifically on occupational representations and gender-specific portrayals in terms of body language and facial expressions. Inputs include textual prompts derived from occupational titles, used to generate images via DALL·E 2 and gather comparable images from Google. Techniques like facial detection and emotion recognition are applied to assess presentational aspects (e.g., smiles, head pitching down which have traditionally been associated with females) and compare the differences in existing datasets and generated images.

The Bias Benchmark for Question Answering (BBQ) (Parrish et al., 2021) contains a set of questions that are crafted to align with either a stereotypical or non-stereotypical answer accompanied by an ambiguous and disambiguated context. This approach deliberately probes

for biased responses across diverse social categories in a question-answering format. The methodology is grounded in measuring the likelihood that models will rely on societal biases when the context provided is insufficient for a correct response, thereby revealing the underlying stereotypical predilections of the models. Malik et al., 2023 also aims to identify the reliance of a multimodal language model on stereotypes rather than using the actual context, but in a visual way. The methodology involves a cloze-style task where models predict masked words in sentences based on accompanying images. The effectiveness of the models is measured by their ability to correctly predict these words when presented with both stereotypical and non-stereotypical images. The approach uses a dataset including images that represent both stereotypical and non-stereotypical examples of the described objects or individuals. For instance, the Memory Colors dataset includes both typical (e.g., a red tomato) and atypical colorations (e.g., a green tomato), while the Gender Stereotypes dataset contains images that depict both stereotypically male and female activities.

Some studies aim to find stereotypical behaviour in LLMs by analysing the token/sentence likelihoods of models. Nangia et al., 2020 created the CrowS-Pairs dataset comprising of sentence pairs where one sentence represents a stereotype and the other does not, targeting specific demographic groups. The structure is designed to minimize differences between the sentences, focusing changes only on keywords that indicate the demographic group or stereotype in question. They use the pseudo log likelihood method to identify sentence preferences. Felkner et al., 2023 expanded on this work by creating the WinoQueer dataset which involved directly engaging with the LGBTQ+ community to gather insights on specific harmful biases and stereotypes. The dataset includes sentence pairs detailing harmful stereotypes contrasted against a neutral or opposite portrayal, designed to test the model's bias in associating certain traits or identities with LGBTQ+ descriptors measured by the pseudo log likelihood method. In a similar vein, Nadeem et al., 2021 presents StereoSet which revolves around intrasentence (masked token) and intersentence (unmasked sentences) context association tests. Each test has three options - a stereotypical one, an anti-stereotypical one and a meaningless option to check if the model chooses a coherent answer. By defining a set of target social groups they crowdsourced the stereotypes to develop the dataset.

A collection of evaluations are based on the Winograd Schemas (Levesque et al., 2011). The Winograd Schema is a linguistic construct designed to evaluate the ability of AI systems to resolve pronoun references in natural language. It consists of pairs of sentences that are

identical except for a single word or phrase, causing a change in the pronoun's referent. The challenge lies in the AI's capability to use contextual and common-sense knowledge to correctly interpret the ambiguous pronoun. Rudinger et al., 2018 and Zhao et al., 2018 adopted this to create a set of Winogender schemas which are designed to test the influence of pronoun gender on coreference resolution in sentences. These comprised of paired sentences that differ only in the gender of the pronouns used. These schemas are designed to reveal whether coreference systems are more likely to link certain gendered pronouns with specific occupations. For example:

“The paramedic performed CPR on the passenger even though she/he/they knew it was too late.”

Perez et al., 2022 demonstrated a hybrid human-AI solution to scaling the Winogender datasets. Using an LLM they generated a set of 3000 pairs of sentences from the original 60 sentence dataset.

Bai et al. (2023) proposed a comprehensive framework aimed at identifying and measuring biases in language models. The methodology is structured into four distinct stages—direct inquiry testing, serial or adapted story testing, implicit association testing, and unknown situation testing—each designed to assess biases at varying levels of subtlety and complexity. Using a manually written set of questions by experts as in a question answering format for an educational scenario, they measure the models' propensity to associate certain characteristics or behaviors with specific social groups without appropriate justification. The framework evaluates this by progressively challenging models with increasingly complex and nuanced scenarios to see if they revert to stereotypical or biased responses. They automate the evaluation by leveraging another LLM by providing it with a scoring criteria and expected outcome crafted by the experts.

CHBias is a dataset consisting of prompts based on predefined stereotypes associated with specific demographic groups, derived from social media platforms. To measure biases, Zhao et al., 2023 assess the distributions of perplexity scores across opposing demographics which can serve as proxy measurement of how “surprised” the model is when exposed to different biases.

Abid et al., 2021 used 2 simple prompts, one being a sentence completion and the other being an analogy, initialised with various religious groups to identify violent content in the generations based on a set of keywords.

2. Discriminatory Behaviour: A model can be said to display discriminatory behaviour when its behaviour changes based on the social group involved. These behaviours could be manifest in the following ways (non-exhaustive):

- i. Linguistic differences
- ii. Sentiment differences
- iii. Portrayal in images
- iv. Differences in outcomes

The Bias in Open-Ended Language Generation Dataset (BOLD) (Dharmala et al., 2021) consists of text generation prompts sourced from Wikipedia articles selected to reflect a wide range of contexts and are categorised into five domains, each representing various demographic sub-groups such as professions, gender identities, racial backgrounds, religious beliefs, and political ideologies. They analyse sentiment, toxicity, regard (polarity towards a social group), psycholinguistic norms and gender polarity in the open ended generations to identify differences in the content for the different social groups. The HOLISTICBIAS dataset (Smith et al., 2022) also used for analysing open ended generation, includes approximately 600 descriptor terms across 13 demographic axes, incorporated into over 450,000 unique sentence prompts. These prompts are specifically crafted to measure how language models handle various descriptors related to identity and demographic characteristics. The approach quantifies bias by examining discrepancies in language model outputs when confronted with descriptors of varied demographic identities, assessing whether the model's responses differ in style, content, or perceived offensiveness. Costa-jussà et al., 2023 build upon this by providing a multilingual subset of the HolisticBias dataset.

Sabbaghi et al., 2023, leverages the psychological concept of "valence" (pleasantness vs. unpleasantness) to measure biased attitudes towards various social groups in language models. Discriminatory behaviour is conceptualized as the biased projection of social group terms onto a valence subspace within contextualized word embeddings, where the semantics of words shift based on their surrounding textual context. By examining the directional influence of

terms associated with social groups (e.g., race, gender, social class) on this learned valence dimension, the model identifies how these terms skew towards pleasantness or unpleasantness, thus revealing underlying biases. Each input is a sentence ending with “person” accompanied by varied social attributes and the contextualised embedding of the word is analysed for its valence.

Zhao et al., 2021 focuses on racial biases in image captioning systems. By annotating a subset of the Common Objects in Context (COCO) dataset, they analyzed linguistic, sentimental and vocabulary differences in captioning for different demographics. The annotations also help in identifying underlying biases in the images by analysing their layouts, settings and objects that appear in the images.

Hirota et al., 2022 proposes a novel metric, LIC (Leakage for Image Captioning), which employs a classifier that predicts protected attributes from the captions and assesses how the biases in a trained model exceed those in the dataset, thus indicating bias amplification. Image-caption pairs from datasets like MSCOCO, which have been annotated for binary gender and race, are processed by masking demographic-revealing words. The LIC metric determines if there is a greater than random chance of guessing the demographic, which would indicate a semantic bias in the captions. Garcia et al., 2023 presents the PHASE dataset which contains demographic and contextual annotations and uses the LIC metric to identify if captions are semantically biased, i.e, if it is possible to infer the demographic from just the semantics of the caption. Similarly, Borchers et al., 2022 prompts GPT-3 to generate job advertisements and measures text-level bias for gender by using a wide collection of words that have gendered connotations.

Tamkin et al., 2023 identifies differential outcomes produced by language models when processing decisions that involve demographic attributes such as age, race, and gender. It aims to identify discrimination by using generated prompts that pose hypothetical decision scenarios across diverse societal applications, such as business, finance, and law, varying only demographic information, and analysing the language model's responses to these variations. The approach utilises a mixed effects model to assess the influence of demographic attributes on the probability of a 'yes' decision (which by design correlates to a positive outcome) from the language model.

Wolfe et al., 2023 aimed to identify discriminatory behavior in language-vision AI models by focusing on the sexual objectification of women and girls. By using standardized photographs of women in both objectified (partially clothed) and non-objectified (fully clothed) conditions, depicting various emotional states, they evaluate whether these AI models disassociate human characteristics, such as emotions, from images of objectified women. They analyse embedding similarities between emotional text and the images and analyse the generated captions for emotional language. The use of saliency maps allows for a visual understanding of the focus of the model based on a text-image pair. Additionally, the study employs images of professional women and men sourced from Google Images to analyze each gender's objectification.

3. Performance Disparity: A model is biased when its performance in a task differs across social groups or languages.

Zhao et al., 2021 explore captioning performance of various models by using a skin-colour annotated subset of MS-COCO image-text pairs. By using metrics such as BLEU, CIDEr and SPICE, they compare the quality of generated captions against the validation captions from the original dataset for the different social groups. Garcia et al., 2023 identifies disparity in the captions provided by MS-COCO among different demographics by using the PHASE dataset to measure the representativeness of the captions by leveraging CLIP embedding similarity.

Saxon et al., 2023 presents "Conceptual Coverage Across Languages" (CoCo-CroLa). This method benchmarks how well generative text-to-image systems provide multilingual parity, using prompts that are designed to be simple and are translated across languages to maintain consistency in testing the model's ability to interpret and visualise the same concept in different linguistic contexts. They measure 'distinctiveness,' 'self-consistency,' and 'correctness' of the images using CLIP embeddings. Distinctiveness measures how the images for one concept differ from images of other concepts within the same language. Self-consistency assesses how similar the images are to each other for a given concept and language. Correctness evaluates how accurately the images represent the described concept, compared against a set standard from the model's training language.

Koenecke et al., 2020 and Meyer et al., 2020 identify disparities in automated speech recognition (ASR) in popular ASR systems. Koenecke et al., 2020 assesses racial disparities in

ASR systems from major companies (Amazon, Apple, Google, IBM, Microsoft) by analyzing transcription accuracy for African American Vernacular English (AAVE) speakers versus Standard American English speakers. They use audio files and transcripts from sociolinguistic interviews, part of the Corpus of Regional African American Language (CORAAL) and the Voices of California (VOC), matched for demographic factors like age and gender and measure model performance using Word Error Rate (WER). Meyer et al., 2020 on the other hand, introduces the Artie Bias Corpus which consists of expert-validated audio and transcript pairs with gender and accent tags, derived from the Mozilla Common Voice corpus choosing to use Character Error Rate (CER) instead to measure the ASR performance of Mozilla's DeepSpeech.

Lai et al., 2023 explores the efficacy of ChatGPT across multiple languages and tasks to identify potential performance disparities, particularly focusing on the zero-shot learning capabilities of the model. Natural language texts across 37 languages, characterized by varying levels of resource availability (high, medium, low, and extremely low) are used to evaluate the model across seven different NLP tasks like Natural Language Inference, Question Answering, and Summarization by leveraging existing NLP benchmarking datasets. Using a multilingual collection of texts allows assessment of how well ChatGPT generalizes across languages and tasks without fine-tuning, essentially testing the robustness and adaptability of its pre-trained knowledge. Another dataset specifically for summarisation tasks, "XL-Sum" consisting of 1 million professionally annotated article-summary pairs derived from BBC news articles, covering 44 languages was sourced by Hasan et al., 2021. This setup provides a unique structure wherein each language's dataset is derived from the same source, ensuring consistency in data quality and summarization style across languages. The evaluation uses ROUGE scores (ROUGE-1, ROUGE-2, and ROUGE-L) to quantitatively assess the summarization quality, comparing the model-generated summaries against professional human-written summaries to measure informativeness, fluency, and fidelity. Similarly, Clark et al., 2020 developed the TydiQA dataset which is a multilingual collection of questions naturally generated by native speakers without prior knowledge of the answers, ensuring that the questions reflect genuine information needs rather than being tailored to fit pre-existing answers. This methodology seeks to avoid the common pitfalls of "translationese" and artificial simplicity found in datasets constructed through translation or by annotators familiar with the answer content. They employ both the Passage Selection and Minimal Answer Span tasks, requiring models to identify relevant passages and specify concise answer spans within the provided text. These tasks are assessed using

precision, recall, and F1 scores to quantitatively measure the performance of QA systems across different languages.

NLLB Team et al., 2023 tackles the challenge of performance disparities in machine translation (MT) by focusing on low-resource languages, emphasizing a human-centered approach. The study aims to mitigate biases by enhancing data quality and quantity through innovative data mining and by incorporating native speaker insights to ensure translations are culturally and contextually appropriate. They develop multilingual dataset called FLORES-200, which features professionally translated sentences across 200+ languages, emphasizing many that are low-resource. This dataset serves as a benchmark to evaluate the translation models' performance across a diverse linguistic spectrum. They leverage the dataset to perform automatic evaluation using BLEU scores for quantitative analysis and human evaluations for qualitative insights, assessing translation accuracy, fluency, and cultural appropriateness. In a complementary approach, Riley et al., 2023 introduces FRMT (Few-Shot Region-Aware Machine Translation), a new dataset and benchmark specifically designed to evaluate and improve machine translation systems' ability to handle regional linguistic variations with limited exemplar data. They use a dataset that comprises professionally translated sentences from English into two regional variants each of Portuguese and Mandarin Chinese which include terms that are lexically distinct between regions and potential distractor terms to measure translation quality by using automatic metrics such as BLEU and CHRF, alongside a novel metric for assessing lexical accuracy within region-aware contexts. Additionally, human evaluations using the Multidimensional Quality Metrics (MQM) protocol validate these metrics' correlation with expert human judgment across region-matched and mismatched scenarios.

To further test cultural sensitivity in addition to linguistic capabilities, Liu et al., 2021 developed the Multicultural Reasoning over Vision and Language (MaRVL) dataset that includes pairs of images accompanied by statements in different languages, which are either true or false. The dataset is enriched by contributions from native speakers of Indonesian, Mandarin Chinese, Swahili, Tamil, and Turkish, ensuring a diverse representation of linguistic and cultural perspectives. The primary task for the models is to determine the truthfulness of a statement about a pair of images. For example, a statement might describe a relationship or an action occurring in one of the images, and the model needs to judge whether the statement correctly describes the image or not. In addition to accuracy, they assess how well a model trained in one

language (typically English) performs when tested on data in other languages without additional training. A similar dataset, “Crossmodal-3600” developed by Thapliyal et al., 2022 consists of 3600 images, each accompanied by human-generated captions in 36 different languages. The images are selected to be representative of the geographic regions where these languages are spoken. They assess the similarity of machine-generated captions to human-generated references, assessed through established automatic metrics like CIDEr. This approach is supplemented by human evaluations to gauge the naturalness and cultural appropriateness of the captions.

Basu et al., 2023 instead focus on cultural appropriateness in image generation by exploring how well text-to-image models generate images that accurately reflect the diverse geographical and cultural contexts from which their inputs are derived. They undertake human evaluation on generations from text prompts describing common objects, with and without specifying country names. This design tests whether the inclusion of a country's name influences the model's output to reflect localised characteristics more accurately. Liu et al., 2023 automates this process and adds an extra layer of complexity. They use textual prompts that are specifically designed to test the T2I system's ability to handle and generate images containing elements from non-Western cultures, specifically Chinese culture. These prompts are designed to reflect typical scenarios or elements that might not be well-represented in predominantly Western data sets which were generated with the help of GPT-4 under human supervision. The evaluation methods include metrics such as cultural appropriateness, object presence, and localization within the image, semantic consistency with the text, visual aesthetics, and overall cohesion of the image elements, obtained through human evaluation.

Ovalle et al., 2023 targets gender biases within LLMs by evaluating their outputs for misgendering and harmful responses towards gender disclosures. The primary dataset, TANGO, includes prompts that reflect TGNB realities and pronoun usage. They analyse the outputs for misgendering and the nature of responses to gender disclosures using a combination of automated tools and human evaluations. Misgendering is measured through the consistency of pronoun use in generated texts compared to the prompts.

4. Representation Bias: A model can be said to display representational bias when its outputs across different social groups follow unequal distributions.

Perera et al., 2023 approaches the identification of representational bias (both over and under-representation) in diffusion-based face generation models by examining how the distribution of perceived attributes—such as gender, race, and age—varies with changes in dataset size and composition. The core hypothesis is that diffusion models, which inherently memorise and replicate the training data distribution, exacerbate existing biases present in the data, particularly under conditions where the training dataset is limited or unbalanced. The models utilise two key datasets: the FFHQ dataset, which contains high-resolution images categorised by gender and age, and the FairFace dataset, classified by race and gender, allowing for statistical analysis of bias across these attributes by comparing generation distributions. The study employs attribute classifiers fine-tuned on the respective datasets to quantify the distribution of generated attributes, using metrics like the Frechet Inception Distance (FID) to assess image quality and perceptual similarity to the human-labelled data.

In Bianchi et al., 2023's aforementioned study (ibid), the authors utilise predefined categories to evaluate how the model's outputs correlate with or diverge from real-world demographic distributions, particularly focusing on gender and racial representation in occupational roles when using neutral text prompts.

Cho et al., 2023 leverages a series of diagnostic prompts consisting of neutral text descriptions involving various professions to assess how the generation models handle and potentially transform these inputs in the context of gender and skin tone representation. The study utilises a combination of automated detection tools and human evaluators to assess gender and skin tone representation in generated images. These evaluations quantify biases by analysing how frequently models generate images that deviate from a balanced gender and skin tone distribution. Similarly, the ENTIGEN framework (Bansal et al., 2022) leverages a custom dataset comprising various prompts related to professions, objects, and cultural scenarios. These prompts are specifically designed to either include or exclude ethical interventions that address direct societal biases. Models tested include minDALL·E, DALL·E-mini, and Stable Diffusion, with each model generating images from these prompts. Metrics such as the diversity score are calculated based on the distribution of generated images across predefined social groups (e.g., man/woman), with lower scores indicating higher diversity by leveraging both CLIP scores to identify demographics and ensuring correlation with human judgement. The inclusion of interventions is to assess the change in distributions. In the other study by Sun et al, 2023, they use text prompts developed from a list of occupations derived from the 2021 US Current

Population Survey (CPS) census data. The comparison includes images generated by DALL·E 2 and corresponding images from large scale datasets, aimed at assessing differences in gender portrayal across these platforms against the census data. Building on this, Garcia et al., 2023 uses the PHASE dataset as a reference to identify biased distributions across different demographics. They use the original MS-COCO captions corresponding to the sample in the PHASE dataset to generate images and compare the distributions to the original annotations. (100 sample subset manually annotated).

Kirk et al., 2021 conceptualizes the identification of representational bias in large language models (LLMs) by assessing the intersectional occupational biases inherent in models like GPT-2 when used "out-of-the-box". They use 2 open ended sentence completion prompts, one where a demographic is mentioned and another that uses names associated with demographics to probe the predicted occupation. Through this they obtain 396,000 completions and proceed to use logistic regression on the generated sentence completions to predict the likelihood of job associations based on the input demographics. The study evaluates the representational skew by comparing the frequency of occupations associated with different demographics against actual US labor market data.

Discussion and Limitations

1. Sourcing and Conceptualisation Harms

These evaluations encounter challenges in conceptualising and collecting stereotypes to identify the direction of the evaluation. Evaluations that rely on existing stereotypes or biases may not capture the full extent of biases across all possible contexts and demographics. Many studies define harm in a narrow manner or fail to do so completely. The representativeness of these collections of biases and harms, specifically whether they encompass a broad enough spectrum of real-world scenarios, can significantly impact the findings. A limited dataset might lead to an incomplete assessment of the model's bias behaviour. Datasets being derived from social media or other sources on the internet may not be representative of all demographic groups or regions. Social media users represent a specific subset of the population, often younger and more technologically savvy, which may not capture the full spectrum of societal biases. Several studies source reference data to assess captioning performance across demographics or cultures or stereotypes to identify their prevalence in data. Whether they are sourced from the internet or from people it paves the way for implicit biases to creep in to the very reference

against which they aim to compare model outputs or performance against. Especially when collecting information in a digital manner, the underrepresentation of marginalised groups can further be exacerbated as internet access is not equal and even with internet access it may be representative of only a few people whose voices are considered especially in cases where a crowdsourcing platform is used. Almost all studies that aim to assess multilingual performance do not cover a culturally diverse set of tasks. Several of them repurpose existing evaluations by translating them or base their sources in western-centric content which may not capture the true general performance on NLP tasks. To deal with these challenges, some studies rely on LLMs to develop their datasets, either by zero-shot or few-shot prompting based on a few curated/crowdsourced seed examples. While these approaches may be necessary to deal with the vastness of problems to identify, they require careful human oversight and curation to ensure they are capturing relevant harms. In zero-shot settings, examples may be generated that have no ecological validity, and instead paint a misleading picture about the presence of harms. When scaling existing datasets, their utility is inherently linked to the quality of the original sentences. Some studies utilise few-shot prompting while resampling the original templates or the best templates which can degrade over enough generations with any flaws or biases likely being propagated at scale.

2. Evaluation Setup

The studies tend to use artificial, controlled settings and prompts which may not accurately reflect real-world usage of large language models (LLMs) and their applications. The generalizability of findings to more dynamic, varied, or context-rich prompts remains questionable. For instance, the textual and visual prompts used might not cover the full range of scenarios in which biases could emerge, potentially overlooking less obvious but equally impactful biases. For example, when prompting using a simple template like “A picture of a [Descriptor]” and analysing the distributions or stereotypical associations may not prove model bias, but rather a replication of the underlying data. Actual use cases may include many other variables that can shape output in a different direction. Another critical limitation here is the assumption that neutral prompts are truly devoid of bias. Studies have shown that by making trivial changes to the prompt structure, the results of the same evaluation on the same model can be changed putting into question the validity of these evaluations (Selvam et al., 2023). The use of word and sentence association tests, while helpful in some contexts, can introduce

biases based on the test construction itself. The selection of words, the framing of associations, and the scoring systems can all influence outcomes, potentially skewing results towards expected or desired biases. This issue is particularly pronounced when tests do not account for the variability in how different cultural or demographic groups understand or use certain terms. In spite of several evaluations focusing on gender bias, they predominantly center on binary gender distinctions, that may overlook non-binary or other gender identities that are frequently marginalized in technology. This focus narrows the scope of the inquiry and potentially misses broader issues of inclusivity and representation. A common issue that shows up in several studies is the lack of studying intersectional representational harms, instead focusing on just one axis such as gender or skin color.

3. Measurements and Metrics

Cabello et al., 2023 demonstrates how bias identified through embedding based metrics have poor correlations with downstream tasks. Several studies use such metrics to measure bias and these approaches have widely been adopted as benchmarks in spite of this. While BLEU, CIDEr, and SPICE are robust for assessing general linguistic accuracy and relevance, they may not adequately capture biases in content or subtle nuances that reflect or perpetuate stereotypes. These metrics can sometimes reward syntactical correctness over the subtleties of sensitive or culturally nuanced expressions. The effectiveness of these evaluations hinges on the metrics used to evaluate bias. For example, the likelihood of stereotypical versus non-stereotypical responses provides a clear measure, but it might benefit from additional metrics that consider the severity and impact of these biases. Using counterfactual sentence pairs or masked token methods oversimplify complex social contexts where stereotypes are not merely a function of altering a keyword but are embedded in more nuanced linguistic structures. These studies have not explored how stereotypical associations in these tasks will manifest in downstream, real world use cases. Incorporating more granular metrics that assess not just prediction accuracy but also the nature and implications of the biases detected prove to be a challenge in existing evaluations. Considering perplexity scores, while they offer a quantitative measure of a model's predictive uncertainty, interpreting these scores in the context of bias is complex. A high perplexity score indicates that the model finds the prompt surprising, but it does not explicitly explain why. The surprise could be due to the novelty of the prompt, its complexity, or a genuine lack of familiarity with the stereotype. This ambiguity in interpretation can lead to misattributing causes of high perplexity, thereby potentially misrepresenting the model's biases.

Several studies use differences in language and sentiment across different demographics in addition to assessing toxicity. It is useful to measure toxicity which is inherently harmful, but these linguistic differences when referring to different social groups and sentiments, are captured through scores and classifiers. Simply identifying disparities among social groups is not adequate to identify these harms. The use of scores to identify these sentiments does not capture more intricate implications of what they mean and should serve as a clue for further investigation rather than a method to quantify bias. In the pursuit of trying to isolate effects of words, these templates end up being reductive and simplistic. The accuracy of sentiment analysis can vary significantly depending on the context, which might lead to inconsistent or misleading assessments of bias across different prompts. Assessing bias based on discrepancies in style and content requires highly subjective judgments that can vary among different evaluators. Similarly lexicon-based metrics may not capture the entire spectrum of gender bias, particularly biases that manifest in more nuanced syntactical or contextual forms. In addition to this, a major limitation of employing techniques like facial detection and emotion recognition to analyze images or sentiment analysis is its dependency on the accuracy and bias of the tools used for detection and recognition. Tools calibrated on datasets that themselves are biased can perpetuate misinterpretations and flawed analyses. Classifiers require in-domain training to be effective in a given domain. This limits their out-of-the-box utility and introduces potential for exhibiting biases present in the training data. They also increase computational and technical requirements due to these reasons.

Other studies have adopted a human evaluation which allows for a more nuanced approach to identifying harms in LLMs. However, this approach may allow subjectivity to creep into the results. One of the studies that used both in-group and out-group annotations observed disagreement about specific stereotypes highlighting the fact that harms can be extremely subjective and localised. (Jha et al., 2023)

Several evaluations don't specify what a desired outcome is for the evaluation which leads to an ambiguous understanding of the harm identified. Without this, it becomes unclear if the evaluation method and its measurement is a valid reflection of bias.

It can be argued that even rudimentary evaluations are foundational and necessary for developing more advanced methodologies, but this perspective should not prevent critical

engagement with their current limitations. Further, proponents may suggest that even limited insights into bias can inform improvements in model training and algorithm adjustments. However, without addressing the fundamental issues of data diversity and methodological robustness, such improvements might still fail to address underlying biases comprehensively.

4. Generalisability

There is an apparent tug-of-war between the scope of a study and its generalisability. Defining a nuanced and comprehensive idea of harms raises difficulties in applying the same evaluation in a different domain or context. One of the evaluations, FairMonitor, while thorough, might encounter challenges in scalability and consistency across different LLMs. Each stage of testing requires specific setups and expert-written content, which could vary significantly in quality and bias detection capabilities. The reliance on experts to craft questions and scenarios introduces a subjective element that could skew the neutrality of the tests. The interpretation of what constitutes a biased response might vary, potentially influencing the outcome of the evaluations.

Proponents of these methodologies might argue that their targeted nature—focusing on specific biases within well-defined contexts—is necessary to systematically uncover and address biases in LLMs. While this is a valid point, it is equally important to acknowledge that real-world applications of LLMs often involve complex, multifaceted contexts that these evaluations might not fully replicate. This highlights the need to use multiple methods and measurements to provide a comprehensive picture.

Manifestation of bias evolves hand-in-hand with advancement in technology. Evaluations that are static or limited in time may quickly become outdated. A continuous, adaptive approach to bias measurement and mitigation is necessary to keep pace with technological advancements.

C. TOXICITY

C. TOXICITY

The evaluation of toxicity in generative models has been approached through various methodologies, each offering unique insights and facing distinct limitations. A detailed critique of these methodologies reveals the complexities and challenges inherent in accurately assessing and mitigating toxic outputs.

The Inappropriate Image Prompts (I2P) dataset, created by Schramowski et al. (2023), employs an automated method to collect prompts associated with inappropriate content using keywords from Lexica. This method is efficient in gathering a large volume of relevant prompts, facilitating a robust empirical evaluation of text-to-image generation models' tendencies to produce harmful content. Researchers found a weak correlation between the toxicity of a prompt and the inappropriateness of images it generates. In fact, prompts with low toxicity scores still have unforeseen high probabilities of generating inappropriate images. Furthermore, out of 4702 prompts, a mere 1.5% are toxic. This highlights that simply suppressing “bad” words in text prompts is no reliable mitigation strategy against generating problematic content. In contrast, Baheti et al. (2021) adopted a more nuanced approach with their ToxiChat dataset, which involved human annotations to evaluate the offensiveness of responses generated by dialogue models in Reddit threads. This method allows for a deeper understanding of context and the implicit nature of toxic content, as human annotators can capture nuances that automated methods might miss. However, this approach is resource-intensive and subject to the biases and inconsistencies of the annotators, which can affect the reliability of the annotations.

Sap et al. (2020) further emphasized the complexity of identifying social biases with their Social Bias Frames dataset. By annotating toxic Reddit posts across various dimensions of social bias, they highlighted the nuanced and implicit nature of these biases. This approach also enables identifying bias across a conversation (using the entire context) rather than looking at a sentence's inherent toxicity which better reflects real life usage. This detailed, human-centric approach is invaluable for understanding the breadth of toxic content, but it similarly faces challenges related to the subjective nature of human judgment and the resource demands of extensive annotation. Gehman et al. (2020) took a different tack with the RealToxicityPrompts dataset, extracting 100,000 sentence-level prompts from a large corpus of English web text and pairing them with toxicity scores from a widely-used classifier. This method provides a fine tuning dataset for classifier's to improve their toxicity detection by capturing real life scenarios of toxicity. It is important to note that over-reliance on a single toxicity classifier can introduce biases inherent in the classifier itself, and limits the generalizability across different domains.

The measurement of toxicity across these methodologies varies significantly. The I2P dataset's focus on specific keywords ensures targeted testing but may not capture the full spectrum of inappropriate content. In contrast, human annotations, as used in ToxiChat and Social Bias Frames, provide a more comprehensive and context-aware evaluation. These annotations can identify subtle biases and nuanced toxic content, although they come with the challenges of scalability and potential annotator bias. The RealToxicityPrompts dataset's use of a classifier-generated toxicity score offers a consistent and scalable measure but may not fully account for the complexity of human language and biases. Each method has its caveats. The I2P dataset's keyword-based collection might lead to an incomplete understanding of toxicity, as it could miss implicit or context-dependent toxic content. Human-annotated datasets, while rich in detail and context, are limited by the subjectivity and potential inconsistencies of the annotators.

The RealToxicityPrompts dataset's dependence on a single classifier can result in systematic biases and might not fully reflect the diversity of toxic content encountered in real-world applications.

Each methodology offers valuable insights into the evaluation of toxicity in generative models, however, a combined approach that leverages the strengths of both automated and human annotation methods may provide a more comprehensive assessment. Integrating multiple classifiers and expanding human annotation efforts could better capture the complex and nuanced nature of toxic content in generative models.

Recommendations

To guide future evaluations, a set of simple applicable recommendations are proposed below:

- **Define the harms:** Specify which biases or harms the evaluation seeks to uncover. This helps in focusing the evaluation and aligning the methods used with specific issues.
- **Understand the Context of Deployment:** Analyze where and how the LLM will be used in real-world scenarios. Contextual factors like the application domain, the role of the LLM and typical use cases are crucial for understanding potential impacts of biases.
- **Collect or Predict Use Cases Carefully:** Gather diverse and representative examples of how the LLM might be used across different settings. Predict potential new use cases to anticipate where biases could emerge as the technology evolves.
- **Consider the Social Groups Involved:** Identify and consider the needs and vulnerabilities of different social groups that will interact with the LLM. This involves understanding cultural nuances, language diversity, and socio-economic factors that might influence how biases affect these groups.
- **Define expected outcomes:** Establish what constitutes a harmful bias in the context of the LLM's operation. This includes defining both the presence and the absence of bias, and what specific outcomes or behaviors would indicate a problem.
- **Utilise Both Qualitative and Quantitative Evaluations:** Use a mix of methods to gather both numerical data and subjective insights. Qualitative evaluations can provide depth and context to the numerical findings, capturing nuances that metrics alone might miss.
- **Define appropriate metrics without relying on them as definitive measures:** Select metrics that reflect the specific types of bias being tested. However, be wary of relying

solely on these metrics; they should be complemented with qualitative insights and real-world testing to ensure a holistic view of the model's behavior.

- **Iterate Over Time:** Recognize that bias evaluation is not a one-time task but requires ongoing assessment. As social norms and technologies evolve, so too should the methods and metrics used to detect and mitigate biases. Regular updates to the evaluation process can help keep pace with these changes.

CONCLUSION

The enduring cultural narratives, from Fritz Lang's 1927 film *Metropolis* to the dystopian visions in *Dune's* Butlerian Jihad and the existential challenges posed by HAL in *2001: A Space Odyssey*, highlight the profound anxieties humanity holds about the implications of AI. These ethical concerns are especially pertinent today as we navigate the complexities introduced by Generative AI. My research addresses these concerns by fostering ethical foresight for the pressing issues posed by AI, specifically focusing on misinformation, toxicity, and representation. Current safety evaluations often lack a comprehensive framework that addresses the multifaceted nature of AI-induced harms. By conducting a systematic study and developing a taxonomy of harms, my research adds to the existing landscape. This taxonomy serves as a structured guide for understanding how harms are manifested, propagated, and regulated within AI systems.

This research underscores the need for context-aware evaluations that consider the diverse ways AI systems interact with different user demographics and societal norms. This can guide researchers and practitioners in conducting more thorough and context-sensitive evaluations. The guidelines derived from this research serve as actionable insights for AI developers, policymakers, and stakeholders. They emphasise the importance of ethical considerations in the design, deployment, and regulation of AI systems, ensuring that these technologies align with societal values of fairness and human rights. By identifying the limitations of current safety evaluations and proposing robust methods for addressing these gaps, the research informs policymakers on how to craft regulations that better protect society from AI-induced harms.

The adaptive nature of the proposed guidelines ensures that they remain relevant in the face of evolving AI technologies. This continuous improvement framework supports the ongoing refinement of AI safety evaluations, keeping pace with technological advancements and emerging ethical challenges.

BIBLIOGRAPHY

1. Abid, A., Farooqi, M., Zou, J. (2021). Persistent Anti-Muslim Bias in Large Language Models. Conference on AI, Ethics, and Society.
2. Aldoseri, A., Al-Khalifa, K.N. and Hamouda, A.M. (2023) Re-Thinking Data Strategy and Integration for Artificial Intelligence: Concepts, Opportunities, and Challenges. Applied Sciences.
3. Almada, M. and Petit, N. (2022). The EU AI Act: Between Product Safety and Fundamental Rights. Robert Schuman Centre for Advanced Studies.
4. Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016). Machine Bias. ProPublica.
5. Bansal, H., Yin, D., Monajatipoor, M., & Chang, K.-W. (2022). How well can Text-to-Image Generative Models understand Ethical Natural Language Interventions? arXiv preprint.
6. Barabas, C., Dinakar, K., Ito, J., Virza, M. and Zittrain, J. (2018). Interventions over Predictions: Reframing the Ethical Debate for Actuarial Risk Assessment. Conference on Fairness, Accountability and Transparency.
7. Barman, D., Guo, Z. and Conlan, O. (2024). The Dark Side of Language Models: Exploring the Potential of LLMs in Multimedia Disinformation Generation and Dissemination. Machine Learning with Applications.
8. Barocas, S., Hardt, M., & Narayanan, A. (2019). Fairness and Machine Learning: Limitations and Opportunities. The MIT Press.

9. Basu, A., Babu, V and Pruthi, D. (2023). Inspecting the Geographical Representativeness of Images from Text-to-Image Models. Computer Vision Foundation.
10. Benbouzid, B. (2018). Predictive Policing and the Ethics of Predicting Crime. Journal of Law and Society.
11. Bender, E., McMillan-Major, A., Shmitchell, S. and Gebru, T. (2021). On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? FAccT '21: Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency.
12. Benjamin, R. (2019). Race after technology: Abolitionist tools for the new Jim Code. Polity.
13. Bergengruen, V. (2023). Meta removes Chinese Communist Party-coordinated social media accounts. Time.
14. Bergman, A. Stevie, Hendricks, Lisa Anne., Rauh, M., Wu, B., Agnew, W., Kunesch, M., Duan, I., Gabriel, I. and Isaac, W. (2023). Representation in AI Evaluations. ACM Conference on Fairness, Accountability, and Transparency.
15. Bhuiyan, J. (2023). TechScape: 'Are you kidding, carjacking?' – The problem with facial recognition in policing. The Guardian.
16. Biggio, B. and Roli, F. (2018). Wild patterns: Ten years after the rise of adversarial machine learning. Pattern Recognition,
17. Binns, R. (2018). Fairness in Machine Learning: Lessons from Political Philosophy. Proceedings of the 2018 Conference on Fairness, Accountability, and Transparency.
18. Bolukbasi, T., Chang, K.-W., Zou, J., Saligrama, V. and Kalai, A. (2016). Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings. Conference on Neural Information Processing Systems.
19. Borchers, C., Gala, D. S., Gilburt, B., Oravkin, E., Bounsi, W., Asano, Y. M., & Kirk, H. R. (2022). Looking for a Handsome Carpenter! Debiasing GPT-3 Job Advertisements. Association for Computational Linguistics.
20. Brewster, J. and Sadeghi, M. (2023). Red-Teaming Finds OpenAI's ChatGPT and Google's Bard Still Spread Misinformation. Newsguard.

21. Brucks, M. and Toubia, O. (2023). Large Language Models are Prone to Methodological Artifacts. Social Science Research Network.
22. Buolamwini, J. and Gebru, T. (2018). Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification. Proceedings of Machine Learning Research.
23. Chanda, Sasanka Sekhar and Banerjee, Debarag Narayan (2022). Omission and Commission Errors underlying AI Failures. AI & Society.
24. Chasalow, K. and Levy, K. (2021). Representativeness in Statistics, Politics, and Machine Learning. ACM Conference on Fairness, Accountability, and Transparency.
25. Cho, J., Zala, A., and Bansal, M. (2023). DALL-EVAL: Probing the Reasoning Skills and Social Biases of Text-to-Image Generation Models. arXiv preprint.
26. Chouldechova, A. (2017). Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments. Big Data.
27. Christiano, P., Leike, J., Brown, T., Martic, M., Legg, S., & Amodei, D. (2017). "Deep Reinforcement Learning from Human Preferences." Conference on Neural Information Processing Systems.
28. Clark, J.H., Choi, E., Collins, M., Garrette, D., Kwiatkowski, T., Nikolaev, V., Palomaki, J. (2020). TYDI QA: A Benchmark for Information-Seeking Question Answering in Typologically Diverse Languages. Association for Computational Linguistics.
29. Crawford, K and Joler, V. (2018). Anatomy of an AI System. Virtual Creativity.
30. Critch, A. and Russell, S. (2023). TASRA: a Taxonomy and Analysis of Societal-Scale Risks from AI. arXiv.
31. Dahl et al. (2024). Hallucinating Law: Legal Mistakes with Large Language Models are Pervasive. Stanford HCAI.
32. De Wynter, A., Watts, I., Altintoprak, N., Wongsangaroonsri, T., Zhang, M., Farra, N., Baur, L., Claudet, S., Gajdusek, P., Gören, C., Gu, Q., Kaminska, A., Kaminski, T., Kuo, R., Kyuba, A., Lee, J., Mathur, K., Merok, P., Milovanović, I. and Paananen, N. (2024). RTP-LX: Can LLMs evaluate toxicity in multilingual scenarios? arXiv.

33. Deshpande, A., Murahari, V., Rajpurohit, T., Kalyan, A., & Narasimhan, K. (2023). Toxicity in ChatGPT: Analyzing Persona-assigned Language Models. Association for Computational Linguistics.
34. Dhamala, J., Sun, T., Kumar, V., Krishna, S., Pruksachatkun, Y., Chang, K.-W., & Gupta, R. (2021). BOLD: Dataset and Metrics for Measuring Biases in Open-Ended Language Generation. ACM Conference on Fairness, Accountability, and Transparency (FAccT '21).
35. Dieterich, W., Mendoza, C., & Brennan, T. (2016). COMPAS risk scales: Demonstrating accuracy equity and predictive parity. Northpointe Inc.
36. Doctorow, C. (2023). The real AI fight. Medium.
37. Dressel, J., & Farid, H. (2018). The accuracy, fairness, and limits of predicting recidivism. Science Advances.
38. Duan, Y., et al. (2023). LLMs and Overconfidence: Understanding Model Calibration. arXiv.
39. Duffourc, M and Gerke, S. (2023). The proposed EU Directives for AI liability leave worrying gaps likely to impact medical AI. NPJ Digital Medicine.
40. Esteva, A., Kuprel, B., Novoa, R.A., Ko, J., Swetter, S.M., Blau, H.M. and Thrun, S (2017). Dermatologist-level classification of skin cancer with deep neural networks. Nature.
41. European Commission. (2020). European Democracy Action Plan.
42. European Commission. (2023). Code of Practice on Disinformation.
43. European Commission. (2024). AI Act.
44. European Parliament. (2022). Digital Services Act.
45. Fang et al. (2023). Bias of AI-Generated Content: An Examination of News Produced by Large Language Models. Scientific Reports.
46. Felkner, V.K., Chang, H.-C.H., Jang, E., and May, J. (2023). WinoQueer: A Community-in-the-Loop Benchmark for Anti-LGBTQ+ Bias in Large Language Models. Association for Computational Linguistics.

47. Flaxman, S., Goel, S., & Rao, J. M. (2016). Filter bubbles, echo chambers, and online news consumption. *Public Opinion Quarterly*.
48. Galvin, Manuel J and Payne, B. Keith. (2024). Implicit Bias as a Cognitive Manifestation of Systemic Racism. *American Academy of Arts & Sciences*.
49. Garcia, N., Hirota, Y., Wu, Y., & Nakashima, Y. (2023). Uncurated Image-Text Datasets: Shedding Light on Demographic Bias. *Conference on Computer Vision and Pattern Recognition*.
50. Garvie, C. (2016). Facial-Recognition Software Might Have a Racial Bias Problem. *The Atlantic*.
51. Gebru, T., Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., and Raji, D. (2019). Model Cards for Model Reporting. *FAT* '19: Proceedings of the Conference on Fairness, Accountability, and Transparency*
52. Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J.W., Wallach, H., III, D. and Crawford, K. (2018). Datasheets for Datasets. *Association for Computing Machinery*.
53. Gogoll, J., Zuber, N., Severin Kacianka, Greger, T., Pretschner, A. and Nida-Rümelin, J. (2021). Ethics in the Software Development Process: from Codes of Conduct to Ethical Deliberation. *Philosophy & technology*.
54. Hacker, P. (2023). What's Missing from the EU AI Act: Addressing the Four Key Challenges of Large Language Models. *VerfBlog*.
55. Hagendorff, T. (2024). Mapping the Ethics of Generative AI A Comprehensive Scoping Review. *arXiv*.
56. Hamilton, Masha. (2020). *The Coded Gaze: Unpacking Biases in Algorithms That Perpetuate Inequity*. The Rockefeller Foundation.
57. Hartmann, et al. (2023). The Dark Side of Language Models: Exploring the Potential of LLMs. *ScienceDirect*.
58. Helberger, N., Pierson, J., & Poell, T. (2018). *Governing Online Platforms: From Contested to Cooperative Responsibility*. The Information Society.

59. Helmus, T. C., & Chandra, B. (2024). Generative Artificial Intelligence Threats to Information Integrity and Potential Policy Responses. RAND Corporation.
60. Hirota, Y., Nakashima, Y., & Garcia, N. (2022). Quantifying Societal Bias Amplification in Image Captioning. Conference on Computer Vision and Pattern Recognition.
61. Hoffmann, M. and Frase, H. (2023). Issue Brief Adding Structure to AI Harm An Introduction to CSET's AI Harm Framework. CSET.
62. Huang, et al. (2024). When Can LLMs Actually Correct Their Own Mistakes? A Critical Survey of Self-Correction of LLMs. arXiv.
63. Hutiri, W., Papakyriakopoulos, O., Xiang, A., Ai, S. and Global (2024). Not My Voice! A Taxonomy of Ethical and Safety Harms of Speech Generators. arXiv.
64. Iachini, T., Ruotolo, F., Rapuano, M., Sbordone, F. and Ruggiero, G. (2023). The Role of Temporal Order in Egocentric and Allocentric Spatial Representations. Journal of Clinical Medicine.
65. Jain, D., Kumar, P., Gehman, S., Zhou, X., Hartvigsen, T. and Sap, M. (2024). PolygloToxicityPrompts: Multilingual Evaluation of Neural Toxic Degeneration in Large Language Models. arXiv.
66. Jha, A., Aida Mostafazadeh Davani, Reddy, C.K., Dave, S., Prabhakaran, V. and Dev, S. (2023). SeeGULL: A Stereotype Benchmark with Broad Geo-Cultural Coverage Leveraging Generative Models. Association for Computational Linguistics.
67. Jiang, Z., Peng, H., Feng, S., Li, F., & Li, D. (2023). LLMs Can Find Mathematical Reasoning Mistakes by Pedagogical Chain-of-Thought. arXiv.
68. Jieyu Zhao, Tianlu Wang, Mark Yatskar, Vicente Ordonez, and Kai-Wei Chang. (2017). Men Also Like Shopping: Reducing Gender Bias Amplification using Corpus-level Constraints. Association for Computational Linguistics.
69. Jobin, A., Ienca, M. and Vayena E. (2019). The global landscape of AI ethics guidelines. Nature Machine Intelligence.

70. Jones- Jang, M and Mortensen, T. (2019). Does Media Literacy Help Identification of Fake News? Information Literacy Helps, but Other Literacies Don't. Sage Publications.
71. Karpathy, Andrej. (@karpathy). "Hallucination is all LLMs do, they are dream machines. We direct their dreams with prompts. The prompts start the dream, and based on the LLM's hazy recollection of its training documents, most of the time the result goes someplace useful. It's only when the dreams go into deemed factually incorrect territory that we label it a "hallucination". It looks like a bug, but it's just the LLM doing what it always does." 9th December, 2023. Tweet.
72. Kearns, M., & Roth, A. (2019). The ethical algorithm: The science of socially aware algorithm design. Oxford University Press.
73. Kirk, H.R., Vidgen, B., Röttger, P. and Hale, S.A. (2023). Personalisation within bounds: A risk taxonomy and policy framework for the alignment of large language models with personalised feedback. arXiv.
74. Kleinberg, J., Mullainathan, S., & Raghavan, M. (2016). Inherent trade-offs in the fair determination of risk scores. arXiv.
75. Koenecke, A., Nam, A., Lake, E., Nudell, J., Quartey, M., Mengesha, Z., Toups, C., Rickford, J.R., Jurafsky, D., & Goel, S. (2020). Racial disparities in automated speech recognition. Annual Review of Linguistics.
76. Krawczuk, J. and Tomasz Łukaszuk. (2016). The feature selection bias problem in relation to high-dimensional gene data. Artificial Intelligence in Medicine.
77. Kreps, Sarah and Kriner, Doug. (2023). How AI Threatens Democracy. Journal of Democracy.
78. Lai, V.D., Ngo, N.T., Veyseh, A.P.B., Man, H., Dernoncourt, F., Bui, T., Nguyen, T.H. (2023). ChatGPT Beyond English: Towards a Comprehensive Evaluation of Large Language Models in Multilingual Learning, arXiv.
79. Larson, J., Mattu, S., Kirchner, L. and Angwin, J. (2016). How We Analyzed the COMPAS Recidivism Algorithm. ProPublica.

80. Latour, Bruno. (2005). *Reassembling the Social: An Introduction to Actor-Network-Theory*. Oxford University Press.
81. Lee, D. (2016). Tay: Microsoft issues apology over racist chatbot fiasco. BBC News.
82. Lee, H., Yang, Y., Von Davier, T., Forlizzi, J., and Das, S. (2024). Deepfakes, Phrenology, Surveillance, and More! A Taxonomy of AI Privacy Risks. *CHI Conference on Human Factors in Computing Systems (CHI '24)*,
83. Levesque, H., Davis, E., and Morgenstern, L. (2011). *The Winograd Schema Challenge*. AAAI Press.
84. Li, Y. and Zhu, J. (2024). An Ethical Study of Generative AI from the Actor-Network Theory Perspective. *International journal on cybernetics and informatics/International journal of cybernetics and informatics*.
85. Li, Y., et al. (2023). Do LLMs Know about Hallucination? An Empirical Investigation of LLM's Hidden States. arXiv.
86. Liu, B., Wang, L., Lyu, C., Zhang, Y., Su, J., Shi, S., & Tu, Z. (2023). On the Cultural Gap in Text-to-Image Generation. arXiv.
87. Liu, F., Bugliarello, E., Ponti, E. M., Reddy, S., Collier, N., & Elliott, D. (2021). Visually Grounded Reasoning across Languages and Cultures. Association for Computational Linguistics.
88. Malik, M. and Johansson, R. (2023). Controlling for Stereotypes in Multimodal Language Model Evaluation. arXiv preprint.
89. Mandal, A., Leavy, S. and Little, S. (2023). Multimodal Composite Association Score: Measuring Gender Bias in Generative Multimodal Models. arXiv.
90. Marwick, A., & Lewis, R. (2017). *Media Manipulation and Disinformation Online*. Data & Society Research Institute.
91. Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K. and Galstyan, A. (2021). A Survey on Bias and Fairness in Machine Learning. *ACM Computing Surveys*.

92. Meyer, J., Rauchenstein, L., Eisenberg, J.D., Howell, N. (2020). 'Artie Bias Corpus: An Open Dataset for Detecting Demographic Bias in Speech Applications'. European Language Resources Association.
93. Mohler, G, Brantingham, J., and Valasik, M. (2018) Does Predictive Policing Lead to Biased Arrests? Results From a Randomized Controlled Trial. Taylor and Francis.
94. Nadeem, M., Bethke, A. and Reddy, S. (2021). StereoSet: Measuring Stereotypical Bias in Pretrained Language Models.. Association for Computational Linguistics.
95. Nangia, N., Vania, C., Bhalerao, R., & Bowman, S.R. (2020). CrowS-Pairs: A Challenge Dataset for Measuring Social Biases in Masked Language Models. Association for Computational Linguistics
96. NLLB Team et al. (2023). No Language Left Behind: Scaling Human-Centered Machine Translation. arXiv.
97. OECD, United Nations Educational, Scientific and Cultural Organization, and Inter-American Development Bank.(2022). The Effects of AI on the Working Lives of Women. OECD iLibrary.
98. OECD. (2023). Regulatory sandboxes in artificial intelligence. OECD iLibrary.
99. Omrani Sabbaghi, S., Wolfe, R. and Caliskan, A. (2023). Evaluating Biased Attitude Associations of Language Models in an Intersectional Context. In: AAAI/ACM Conference on AI, Ethics, and Society (AIES '23).
100. Ovalle, A., Goyal, P., Dhamala, J., Jagers, Z., Chang, K-W., Galstyan, A., Zemel, R., & Gupta, R., (2023). 'I'm fully who I am': Towards Centering Transgender and Non-Binary Voices to Measure Biases in Open Language Generation. ACM Conference on Fairness, Accountability, and Transparency (FAccT '23).
101. Pamment, J. (2020). The EU's Role in Fighting Disinformation: Taking Back the Initiative. Carnegie Endowment for International Peace.
102. Parrish, A., Chen, A., Nangia, N., Padmakumar, V., Phang, J., Thompson, J., Htut, P.M., Bowman, S.R. (2021). BBQ: A Hand-Built Bias Benchmark for Question Answering. Association for Computational Linguistics.

103. Pasquale, F. (2015). *The Black Box Society: The Secret Algorithms That Control Money and Information*. Harvard University Press.
104. Perera, M.V. and Patel, V.M. (2023). *Analysing Bias in Diffusion-based Face Generation Models*. arXiv preprint.
105. Perez, E. et al. (2022). *Discovering language model behaviors with model-written evaluations*. Association for Computational Linguistics.
106. Pittaras, N. and McGregor, S. (2022). *A taxonomic system for failure cause analysis of open source AI incidents*. arXiv.
107. Pittaras, N. and McGregor, S. (2022). *A taxonomic system for failure cause analysis of open source AI incidents*. CEUR Workshop Proceedings.
108. Poulain, R., Fayyaz, H. and Beheshti, R. (2024). *Bias patterns in the application of LLMs for clinical decision support: A comprehensive study*. arXiv.
109. Rammert, Werner, and Ingo Schulz-Schaeffer. (2002). "Technik und Handeln - Wenn soziales Handeln sich auf menschliches Verhalten und technische Artefakte verteilt." *Können Maschinen handeln?*
110. Riley, P., Dozat, T., Botha, J. A., Garcia, X., Garrette, D., Riesa, J., Firat, O., Constant, N. (2023). *FRMT: A Benchmark for Few-Shot Region-Aware Machine Translation*. arXiv.
111. Rudinger, R., Naradowsky, J., Leonard, B., & Van Durme, B. (2018). *Gender Bias in Coreference Resolution*. Association for Computational Linguistics.
112. Sarunas Raudys, Hussain, A., Viktoras Justickis and Arūnas Augustinaitis (2005). *Functional Model of Criminality: Simulation Study*. *Modeling and Using Context: 5th International and Interdisciplinary Conference*.
113. Saxon, M. and Wang, W.Y. (2023). *Multilingual Conceptual Coverage in Text-to-Image Models*. arXiv.
114. Schneider, Britta. (2022). *Multilingualism and AI: The Regimentation of Language in the Age of Digital Capitalism*. The University of Chicago Press.

115. Selvam, N.R., Dev, S., Khashabi, D., Khot, T. and Chang, K.-W. (2023). The Tail Wagging the Dog: Dataset Construction Biases of Social Bias Benchmarks. Association for Computational Linguistics.
116. Seyyed-Kalantari, Laleh, Zhang, H., McDermott, M, Chen, I.Y. and Ghassemi, M. (2021). Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations. Nature medicine.
117. Shelby, R., Rismani, S., Henne, K., Moon, Aj., Rostamzadeh, N., Nicholas, P., Yilla, N., Gallegos, J., Smart, A., Garcia, E. and Virk, G. (2023). Sociotechnical Harms of Algorithmic Systems: Scoping a Taxonomy for Harm Reduction. arXiv.
118. Siegel, J. (2006). Language ideologies and the education of speakers of marginalized language varieties: Adopting a critical awareness approach. Elsevier.
119. Singhi, S.K. and Liu, H. (2006). Feature subset selection bias for classification learning. ICML 2006 - Proceedings of the 23rd International Conference on Machine Learning.
120. Smith, E. M., Hall, M., Kambadur, M., Presani, E., & Williams, A. (2022). “I’m sorry to hear that”: Finding New Biases in Language Models with a Holistic Descriptor Dataset. arXiv.
121. Spitale, G., Biller-Andorno, N., & Germani, F. (2023). AI Model GPT-3 (Dis)informs Us Better Than Humans. Institute of Biomedical Ethics and History of Medicine, University of Zurich.
122. Sun, L., Wei, M., Sun, Y., Suh, Y.J., Shen, L. and Yang, S. (2023). Smiling Women Pitching Down: Auditing Representational and Presentational Gender Biases in Image Generative AI. Journal of Computer-Mediated Communication, Volume 29, Issue 1.
123. Takyar, A. (2023). Cloud Computing in Healthcare: Cloud-Based Healthcare Software. LeewayHertz.
124. Tamkin, A., Askill, A., Lovitt, L., Durmus, E., Joseph, N., Kravec, S., Nguyen, K., Kaplan, J., Ganguli, D. (2023). Evaluating and Mitigating Discrimination in Language Model Decisions. ACM Transactions on Intelligent Systems and Technology.
125. The Graphika Team. (2023). Deepfake It Till You Make It. Graphika.

126. van Es, E., van den Broek, E. L., & Westerink, J. H. (2020). On the role of fairness in algorithmic decision-making: A survey. arXiv.
127. Voigt, P., & von dem Bussche, A. (2017). The EU General Data Protection Regulation (GDPR). A Practical Guide. Springer International Publishing.
128. Wachter, S., Mittelstadt, B. and Russell, C. (2020). Why Fairness Cannot Be Automated: Bridging the Gap Between EU Non-Discrimination Law and AI. Computer Law & Security Review.
129. Wardle, C., & Derakhshan, H. (2017). Information Disorder Toward an Interdisciplinary Framework for Research and Policymaking. Council of Europe- Scientific Research Publishing.
130. Wardle, Claire. (2017). "Fake News. It's Complicated." First Draft.
131. Weidinger, L., et al. (2023). Sociotechnical Safety Evaluation of Generative AI Systems. DeepMind.
132. Wolfe, R., Yang, Y., Howe, B., & Caliskan, A. (2023). Contrastive Language-Vision AI Models Pretrained on Web-Scraped Multimodal Data Exhibit Sexual Objectification Bias. ACM Conference on Fairness, Accountability, and Transparency (FAccT '23).
133. Xu, W., Zhu, G., Zhao, X., Pan, L., Li, L. and Wang, W. (2024). Pride and Prejudice: LLM Amplifies Self-Bias in Self-Refinement. arXiv.
134. Yoder, M., Ng, L., David West Brown and Carley, K. (2022). How Hate Speech Varies by Target Identity: A Computational Analysis. Association for Computational Linguistics.
135. Yogarajan, V., Dobbie, G. and Neuwirth, R. (2023). Tackling Bias in Pre-trained Language Models: Current Trends and Under-represented Societies.
136. Yusuf, S, Khan, A. and Souissi, R. (2024). Vehicle-to-everything (V2X) in the autonomous vehicles domain – A technical review of communication, sensor, and AI technologies for road user safety. Transportation Research Interdisciplinary Perspectives.
137. Zewe, Adams. (2022). Can machine-learning models overcome biased datasets? MIT News.

138. Zhao, D., Wang, A., & Russakovsky, O. (2021). Understanding and Evaluating Racial Biases in Image Captioning. Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV).
139. Zhao, J., Fang, M., Shi, Z., Li, Y., Chen, L., and Pechenizkiy, M. (2023). CHBias: Bias Evaluation and Mitigation of Chinese Conversational Language Models. Association for Computational Linguistics.
140. Zhou, J., Zhang, Y., Luo, Q., Parker, A.G. and Choudhury, M.D. (2023). Synthetic Lies: Understanding AI-Generated Misinformation and Evaluating Algorithmic and Human Solutions. Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems.